



Review article

Recent advances in 3D models of the nervous system for neural regeneration research and drug development

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ABSTRACT

The development of drugs for nervous diseases poses distinctive difficulties owing to the incomplete understanding of the physiology and complex pathogenesis of the multifaceted central (CNS) and peripheral (PNS) nervous systems. Conventional animal tests and *in vitro* two-dimensional (2D) cell cultures fail to reproduce the sophisticated structure of natural human tissues, hindering the new drug discovery process. The emerging three-dimensional (3D) neural tissue models, including organoids, organ-on-chip and 3D-printed neural scaffolds, can provide an improved reproduction of the critical features, structural complexity, biological functions, dynamic circulation micro-environment and cell-matrix/cell interactions of the nervous systems. This review examines state-of-the-art 3D models for neural physiology/pathology, emphasizing their drug development applications. Fundamental advantages of various *in vitro* 3D neural models for investigating the mechanisms of nerve regeneration and disorders in both the CNS and PNS are compared in terms of the different modeling techniques. In addition, the applications of 3D neural models in drug development are summarized covering a range of areas such as disease modeling for basic research, pharmacokinetic and pharmacodynamic testing for drug screening and drug safety evaluation. Furthermore, current challenges and future outlook of biomimetic models and the existing bottlenecks hindering their successful translation into clinical use are discussed.

Statement of significance: This review highlights the groundbreaking potential of 3D neural models—organoids, organ-on-chip, and 3D-printed scaffolds—to revolutionize neurological research and drug development. Unlike conventional methods, these models replicate the intricate structure and function of the human nervous system, enabling precise study of diseases like Alzheimer's, spinal injuries, and brain tumors. By summarizing and discussing recent advancements, the review compares techniques, their applications in drug screening and personalized medicine, and addresses challenges in model accuracy and scalability. Bridging neuroscience, engineering, and pharmacology, this work provides a roadmap for researchers to innovate therapies. The insights presented are critical for accelerating drug discovery and improving treatment outcomes, making 3D neural models essential for scientists and clinicians tackling neurological disorders.

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Abbreviations

2D	Two-dimensional
3D	Three-dimensional
ABI	Ascl1, Brn3b/3a, and Isl1
AD	Alzheimer's disease
ADME	Absorption, distribution, metabolism, and excretion
AI	Artificial intelligence
ALS	Amyotrophic lateral sclerosis
APOE	Apolipoprotein E
AQP4	Aquaporin-4
A β	Amyloid β -protein
BACE	Beta-secretase
BBB	Blood-brain barrier
BoB	Büngner
BMSCs	Bone marrow mesenchymal stem cells
Cas9	CRISPR-associated protein 9
CB	Carbon black
CD31	Platelet endothelial cell adhesion molecule-1
CNS	Central nervous system
CNS-PNET	CNS primitive neuroectodermal tumor
COs	Cerebral organoids
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
CSF	Cerebrospinal fluid
DNA	Deoxyribonucleic acid
DNAJC6	DnaJ Heat Shock Protein Family (Hsp40) Member C6
DOX	Doxorubicin
ECM	Extracellular microenvironment
FDA	Food and Drug Administration
GAGs	Glycosaminoglycans
gBS	Glioblastoma brain sphere
GLICO	Cerebral organoid glioma
GMLT	Gray matter-like tissue
GRIN2A	Glutamate ionotropic receptor NMDA type subunit 2A
GRIN2B	Glutamate ionotropic receptor NMDA type subunit 2B
GSCs	Glioma stem cells
GSK3 α/β	Glycogen synthase kinase-3 alpha/beta
HAs	Human astrocytes
HCs	Hair cells
HD	Huntington's disease
hESC	Human embryonic stem cell
hiPSCs	Human induced pluripotent stem cells
hNSCs	Human neural stem cells
hPSC	Human pluripotent stem cells
HT	high-throughput
Inh2	Inhibitor 2
iPSCs	Induced pluripotent stem cells
iSG	Induced sensory ganglion
LPC	Lysophosphatidylcholine
MBP	Myelin basic protein
MEA	Multi-electrode array
MG	Myasthenia gravis
microTMA	Micro thermomechanical analysis
miR-7	Microrna-7
mESC	mouse embryonic stem cell
MNs	Motor neurons
MTT	Methylthiazolyldiphenyl-tetrazolium bromide
NAD	Nicotinamide adenine dinucleotide
NCCs	Neural crest cells
NDDs	Neurodegenerative diseases
neoCOR	Neoplastic cerebral organoid
NGF	Nerve growth factor
NMD	N-methyl-D-aspartic acid
NMJs	Neuromuscular junctions
NMOs	Neuromuscular organoids
NV chip	Neurovasculature-on-a-chip

NVU	Neurovascular unit
OoC	Organ-on-a-chip
OPCs	Oligodendrocyte progenitor cells
PCL	Poly- ϵ -caprolactone
PCSM	Porous chitosan microspheres
PD	Parkinson's disease
PDMS	Polydimethylsiloxane
PEGDA	Poly(ethylene) glycol diacrylate
PLGA	Poly-lactic-co-glycolic acid
PNS	Peripheral nervous system
p-Tau	Phosphorylated tau
PU	Polyurethane
QDs	Quantum dots
RSCs	Rat Schwann cells
SAG	Shh agonist
SCI	Spinal cord injury
SCs	Schwann cells
SGNs	Spiral ganglion neurons
SL	Stereolithography
sNPCs	Spinal neuronal progenitor cells
SOD1	Superoxide dismutase 1 Gene
SOX10	SRY-Box Transcription Factor 10
TAR DNA	Transactive response DNA
TBI	Traumatic brain injury
TMMSCs	Tubercular meshwork mesenchymal stem/stromal cells
TMZ	Temozolomide
tPA	Tissue plasminogen activator
vGLUT	Vesicular glutamate transporters
WHO	World Health Organization
WMLT	White matter-like tissue
WT	Wild-type
Z-VAD-FMK	Carbobenzoxy-valyl-alanyl-aspartyl- [O-methyl]-fluoromethylketone

1. Introduction

The human nervous system is both complex and sophisticated, orchestrating vital bodily functions through the intricate interplay of electrical and chemical signals. The central nervous system (CNS) encompasses the brain and spinal cord, and the peripheral nervous system (PNS) plays a fundamental role in maintaining human health [1]. The spectrum of neurological disorders looms large, presenting a formidable challenge to public health. According to World Health Organization (WHO) data, these disorders have emerged as a leading cause of mortality, surpassing even cancer and endocrine conditions since the 1990s [2]. Numerous afflictions, including acute traumatic injuries like spinal cord (SCI) and traumatic brain injuries (TBI), neurodegenerative diseases such as Alzheimer's and Parkinson's, and peripheral neuropathies as a result of autoimmune disorders or infections, affect millions of patients worldwide. Despite concerted efforts to unravel their specific characteristics and devise effective treatments, formidable hurdles persist. Failure rates of new drug research and development are high in late-stage clinical trials compared with other diseases, mainly because of the intricacy of the human nervous system and the challenge of directly observing it *in vivo* [3]. Moreover, the pathogenic aspects of neurological disorders and neurogenetic mechanisms on a molecular level remain unclear, which greatly hinders translational neuroscience research. Normal neuronal cell studies in 2D have been widely applied to investigate the detailed mechanisms of neuronal damage and to screen new pharmacological agents. However, conventional 2D neuronal models offer limited insight into the intricate mechanisms of neural damage, failing to capture the complexities of *in vivo* neurotoxicity [4]. In addition, financial risks for neurodrug development have discouraged new investment from pharmaceutical companies globally [5]. It is also well-known that the limited efficacy of animal models in predicting human conditions is due to species-specific differences. For example, a

human superoxide dismutase (SOD1) gene-engineered mouse is commonly used as an amyotrophic lateral sclerosis (ALS) model [6]. However, the pharmacological compounds recognized through the animal model have been demonstrated to be of limited utility in humans.

In recent years, a paradigm shift has occurred with the advent of 3D culture systems such as organoids [7–9], organ-on-chip [10,11] and other scaffold-based platforms [12–14]. With the ability to integrate diverse cell types, materials and various signals (physical, chemical, electrical, etc.), those 3D models can generate a more complex environment and provide neural tissue-mimic structures *in vitro*, which tend to be more informative and predictive than standard animal models. Driven by different concepts, different 3D models are designed and used with distinct priorities and specialties. Organoids typically use stem cells (pluripotent, fetal or adult) to form self-organized 3D tissue, and are superior in mimicking the biological complexity and early developmental details of an organ [15]. On the other hand, organ-on-chip approaches take the form of microfluidic devices fabricated by micro-manufacturing technology and provide dynamic platforms for guiding minute volumes of liquid passing through the miniature tissues residing on the chips [16]. Both single and multiple organ-like cells or tissues can

be integrated into one device with an engineered biomimetic physiochemical environment to replicate the key structures, functions and interactions among different organs. Recirculating flow can be realized on this microsystem which mimics the circulatory system in the human body to transport nutrients and waste [17]. In addition, engineering-based 3D scaffolds are also constructed by advanced manufacturing technology, such as 3D printing, with optimal mechanical and degradability properties and precise cell arrangement to achieve desired tissue modeling [18]. Those 3D tissue models can play significant roles in not only drug screening but also drug discovery and design, which increases the efficiency of drug development on a large scale. More importantly, by using particular cell organs from specific patients in 3D tissue models, personalized and precision medicine can be expedited. It should be noted that OoCs or scaffolds where cells are cultured on patterned surfaces (e.g., BBB chips with endothelial monolayers, or fibrous structures with seeded cells) without full volumetric integration can be classified as 2.5D systems in the strict sense [19]. These systems are still discussed in this review due to their non-negligible capability to replicate physiological characteristics for constructing *in vitro* neural systems and for therapeutic development. 3D

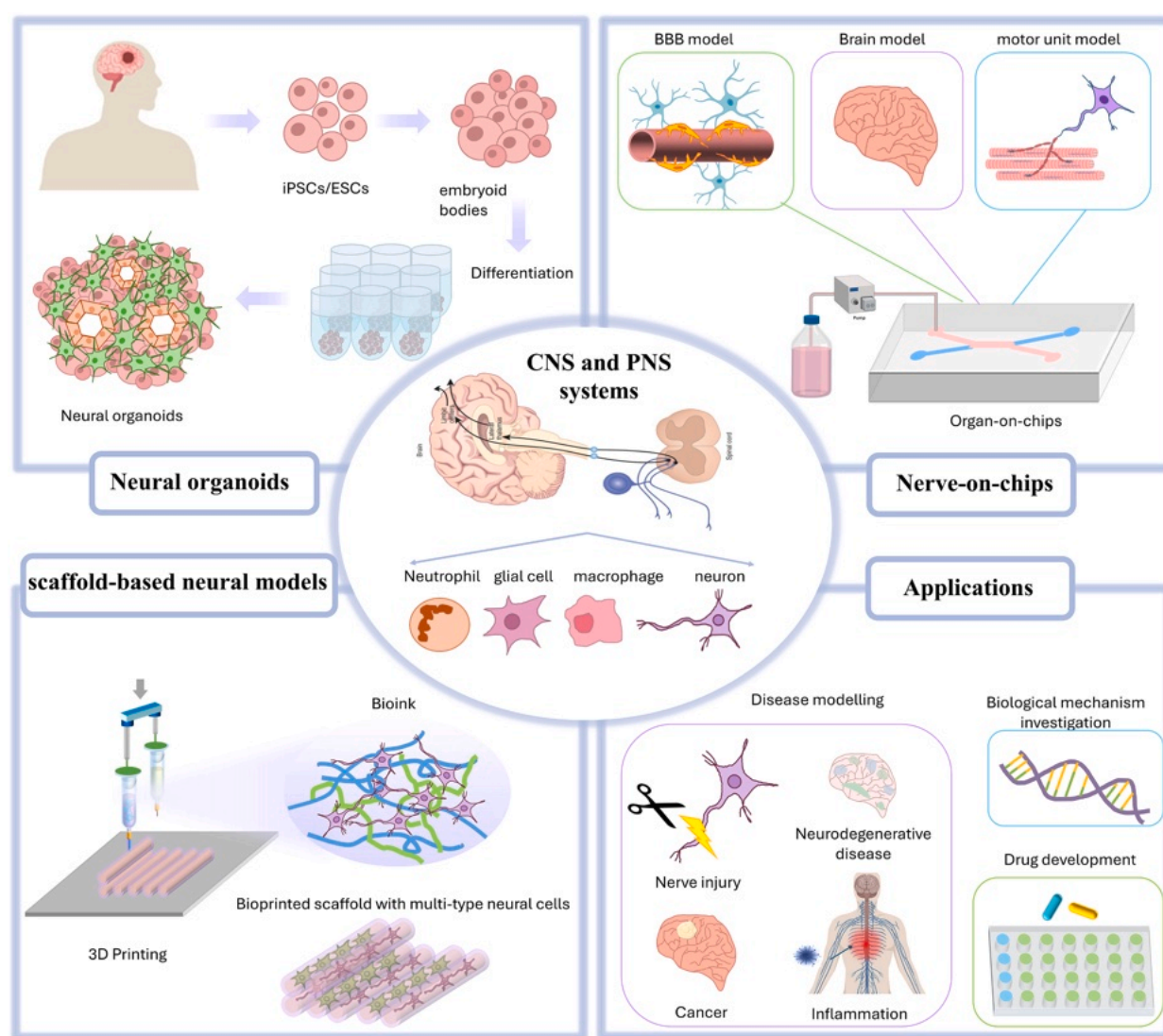


Fig. 1. Schematic illustration of the emerging 3D CNS/PNS models including organoids, organ-on-chip, 3D-printed scaffolds and their applications on neural regeneration and drug development. Image top left was reproduced from [9], copyright 2025, Elsevier. The middle image was reproduced from [1], copyright 2025, Springer Nature. Image bottom right was reproduced from [14] under the terms of the Creative Commons CC BY license.

OoCs based on 3D cell models, including spheroids/aggregates, organoids, multilayers of cells and hydrogels encapsulating cells, as well as 3D bioprinted scaffolds, are certainly involved in the following discussion.

This review endeavors to explore recent developments in 3D *in vitro* neural modeling and their possibility to tackle the challenges of treating neurological diseases (Fig. 1). We cover approaches put forward for the modeling of both the CNS and PNS using organoids, nerve-on-chip, and scaffold-based neural models, summarizing technical methodologies, cell-sourcing strategies, and key functionalities. Furthermore, we review the application of these 3D models in drug discovery and screening, highlighting the advantages of such models for more cost-effective and efficacious therapeutic interventions. Finally, we anticipate future prospects and technical hurdles that must be addressed for the integration of 3D *in vitro* modeling technologies with drug development endeavors. It is expected that through interdisciplinary collaboration and technological innovation, 3D *in vitro* models will represent significant advances in alleviating the burden of neurological diseases, leading to improved patient outcomes.

2. CNS in-vitro models for neural regeneration and disease treatment

2.1. Requirements for *in vitro* CNS models

Adequate CNS *in vitro* models play key roles in replicating and providing information regarding behavioral, functional or systemic responses to the applied signals, which can improve the success of CNS pharmaceutical development [20]. To fully capture the complex physiology, anatomy, and functionality of the CNS, it is important to clarify the requirements that such a system should satisfy.

The first consideration is that the CNS consists of a complex cellular network that includes a diverse array of cell types, such as neurons, astrocytes, oligodendrocytes, and vascular endothelial cells, all embedded within an extracellular microenvironment. Some significant interactions and functions that need to be rebuilt *in vitro* should involve specific cellular types based on the purpose of the 3D model and the disease type to be studied. Neurons are the core cells of the CNS model, and neurons with different functions need to be included in the construction of the CNS model to simulate the complex functions of the nervous system. In the model simulating the function of the cerebral cortex, excitatory pyramidal neurons and inhibitory interneurons are needed to reproduce the neural circuit activity of the cerebral cortex. Astrocytes are involved in maintaining the stability of the microenvironment of neurons, and oligodendrocytes are responsible for the formation of myelin. These glial cells need to be included in the model to accurately simulate the physiological functions of the central nervous system. Vascular endothelial cells participate in the formation of the blood-brain barrier (BBB), which is essential for maintaining the stability of the internal environment of the nervous system. When establishing a model that simulates the physiological or pathological state of the central nervous system (such as vascular diseases like stroke), vascular endothelial cells and the blood-brain barrier structure they form are important research objects. The inclusion of vascular endothelial cells in the model can more realistically simulate the blood supply and material exchange process of the central nervous system, as well as the destruction mechanism of the blood-brain barrier in the disease state.

Another challenge to reproduce *in vitro* is the distinct microenvironment of the CNS composed of multiple physiochemical factors and mechanical properties including complex molecular environment, extremely soft extracellular matrix (ECM), specific cellular architecture, etc. [21]. For example, it is desired but challenging to rebuild a biomimetic micro-environment *in vitro* for CNS with various proteins and molecules (e.g. glycosaminoglycans (GAGs), proteoglycans, etc.) surrounding neural cells, because these molecules play unique roles in

maintaining the steady state of the nervous system, such as regulating specific oxygen levels and specific nutrient compositions [22]. Also, it is vital to tune the mechanical properties of ECM as they are significantly relevant to cell fate and guide cell differentiation [23], while it is difficult to construct *in vitro* models with desired physiological stiffness and viscoelasticity, which simultaneously support cell functions due to the very soft structures with extremely low stiffness of CNS (brain~50 Pa, spinal cord~500–1500 Pa) [24]. In addition, the CNS model needs to have a cell structure similar to that *in vivo*, including the length and branching of dendrites and axons of neurons and the specific morphology of glial cells, which will affect the synaptic connections between neurons and the structure and function of neural circuits. Fortunately, advanced manufacturing techniques such as bioprinting can be adopted to achieve the precise distribution of distinct cellular and molecular varieties in 3D, paving a new way for the creation of neuronal cultures with controlled 3D patterns and directions that enable the investigation of the dynamics and connections in neural networks.

Furthermore, other factors, such as subunit integration and organ-organ interplays, need to be taken into account as well. Despite the mentioned huge challenges that make it complicated to remold the exact conditions of CNS *in vitro*, great advancements have been achieved in recent years indicating that it is possible to reflect body-level CNS disease state enabling prediction of drug response considering individual differences.

2.2. Microphysiological and pathological organoids of CNS

Organoids are self-organized 3D aggregates with multiple cells derived from stem cell differentiation, which mimic the structural features, biological functions and cellular interactions found in natural tissues [25]. By incorporating genetically engineered induced pluripotent stem cells (iPSCs) or patient-derived cells into organoids, phenotypic manifestations at the organ level can be reproduced *in vitro* by expressing specific molecular lesions. As a result, established diseased organoids have the potential to improve the testing of the effectiveness and accuracy of pharmaceutical compounds and can be considered to replace traditional animal models. CNS organoids epitomize a potent instrument for advancing our understanding of CNS development, disease pathology, and therapeutic interventions [26]. Their capacity to replicate fundamental characteristics of the developing brain and spinal cord and model neurological disorders are key features that facilitate drug discovery efforts holding great promise for addressing unmet clinical needs in neurology and psychiatry.

2.2.1. CNS organoids for neural development

Specific mechanisms of human neural development and evolution are expected to be dissected by brain and spinal cord organoids, which have emerged as powerful methods in recent years for modeling cellular and developmental complexity [27,28]. For instance, characteristics that arise in the period following birth can be sufficiently acquired by brain organoids. Gordon et al. developed a cortical organoid with glutamatergic projection neurons that mimicked the natural progression of cortical development during the perinatal period, indicating a transition in the glutamate NMD receptor subunits, shifting from GRIN2B to GRIN2A [29]. However, without blood vessels, the organoids cannot get nutrients deep inside, which limits their size and maturity. In another study, precisely engineered and reproducible vasculature with the capability of perfusion was incorporated into the cortical organoid platform [30]. It was found that human cortical organoid cultured within the vascular scaffold was supported without hindering neuroectodermal differentiation. As a robust, reproducible, and tunable, this scaffold could address some biological questions related to human brain development. To further replicate human brain complexity, neuronal networks with complex patterns of activity were constructed in some cortical organoids (e.g., layered cortical architecture, or glial support), which could open the door to the investigation of sophisticated functions

and formation mechanisms of the brain [31–33].

On the other hand, the spatial orientation and alterations in the structure of planar cells amidst the closure of the neural tube can be assessed by spinal cord organoids, which are vital for examining the initial stages of neuropathogenesis. For example, a patient-derived spinal cord organoid was developed by Winanto et al. to investigate maternally inherited mitochondrial DNA diseases [34]. It was found that neurogenesis and neurite outgrowth (neurite length changes from $\sim 800\ \mu\text{m}$ to $\sim 200\ \mu\text{m}$) were restricted by elevated levels of Notch signalling in the disease model compared to its isogenically corrected counterpart, which might cause the pathogenesis difference.

The construction of spinal cord organoids *in vitro* holds considerable importance for targeted repair and drug screening of spinal cord injury. Xu et al. converted human astrocytes directly into early neuroectodermal cells, where initiation of signalling pathways facilitated by fibroblast growth factor, sonic hedgehog, and bone morphogenetic protein 4 induced their formation of spinal cord organoids [7]. Human astrocyte-derived spinal cord organoid grafts can differentiate into spinal cord neurons in mice experiencing total spinal cord injury, which migrate and form synaptic connections with host neurons, providing a new path for spinal cord injury repair. To further mimic the major morphological and functional properties of both white and gray matters of the spinal cord, Lai et al. developed a tissue engineering method utilizing neural stem cells to modularly assemble a white matter-like tissue (WMLT) module and grey matter-like tissue (GMLT) module of neural stem cells to simulate the white matter and grey matter of the spinal cord and constructed engineered transplantable spinal cord organoids *in vitro* [35]. Cortical motor-evoked potential amplitude and latency were shortened in the injured spinal cord by transplanting the spinal cord organoids, and the WMLT and GMLT regions had targeted repair effects on the white and grey matter constituents of the damaged spinal cord. However, organoids generated from these methods could not completely establish spatial dorsoventral cytoarchitecture in the human spinal cord. Besides, the center of 3D spinal cord organoid tended to be necrosis due to a lack of nutrients with bigger size. In another approach, Xue and colleagues combined porous chitosan microspheres (PCSM) and Matrigel@ Shh agonist (SAG) to form composite scaffolds with human spinal cord organoids to mimic the organizing center of developmental neural tube *in vitro* [36]. Due to the continuous release of SAG, known for its role in ventralizing signalling within the spinal cord, this organoid demonstrated the capability to develop dorsoventral spinal cord-like cellular arrangements, featuring region-specific progenitor cells and neurons, and exhibiting functional calcium activity with important implications in spinal cord injury repair and drug screening.

2.2.2. CNS organoids for neuropsychiatric diseases

Neuropsychiatric diseases including depression, schizophrenia, and autistic spectrum disorder significantly influence various facets of human personality, cognition and perception [37]. It is difficult to investigate the mechanism of depressive disorder experimentally because of the scarce availability of diagnostic indicators or research paradigms since the symptoms and etiology have great heterogeneity. Moreover, there are huge complexities in symptomatology, and poly-genetic etiology [38]. By using cerebral organoids, the neuropsychiatric disease can be modeled with primary cells from patients with developmental abnormalities and genetic alterations. In recent studies, whole-brain organoids were constructed using hiPSCs from patients with schizophrenia, and marked differences were found in the genes related to mitochondrial function using transcriptome sequencing analysis [39]. Besides, the oxygen consumption level was reduced evidently in the schizophrenic organoids by the Seahorse Mito Stress test, and decreased responses to the KCl depolarization or electrical stimulation showed obvious neuronal dysfunction. Many AD patients suffer from neuropsychiatric symptoms (NPS) which are related to dysfunctional serotonergic systems. Alejandro et al. generated hindbrain organoids containing serotonergic neurons by reprogramming human

peripheral blood mononuclear cells into human induced Pluripotent Stem Cells and then differentiating into organoids. They used organoids generated from different patients to assess the treatment effects of different concentrations of escitalopram oxalate, showing that changes in the level of serotonin (5-HT) before and after were dependent on the dose of escitalopram oxalate [40]. While organoids offer unprecedented opportunities to study neuropsychiatric diseases, factors like stress, infection, or diet contribute to these diseases but are difficult to model *in vitro*. Besides, behavioral symptoms (e.g., cognitive deficits) cannot be directly measured, and electrophysiology or omics data may not fully reflect disease phenotypes.

2.2.3. CNS organoids for neurological diseases

The gradual reduction of specifically susceptible neuronal populations, protein misfolding, genetics and ageing are the main causes of neurodegenerative diseases (NDDs) such as Parkinson's disease (PD), Alzheimer's disease (AD), ALS, etc. NDDs pose a big threat to normal human activity and the typical clinical manifestations include cognitive dysfunctions and abnormal behaviors [38]. Building organoids representing the brain structure helps understand the pathogenesis of NDDs and thus may promote the development of effective treatment programs and drugs.

For example, sensorimotor ALS organoids were established replicating physiologically functional neuromuscular junctions (NMJs). Both familial ALS-edited isogenic lines and iPSC lines generated from ALS patients were involved in the organoids, showing impairment of NMJs [41]. Furthermore, since ALS as a neuromuscular disorder is caused by disorders of the NMJs, spinal cord organoids that can involve spinal cord neurons and skeletal muscles are more frequently used. Martins et al. generated neuromuscular organoids (NMOs) by adopting hPSC-derived axial stem cells. Spinal cord neurons and skeletal muscle cells could be simultaneously generated to track the trajectories of neural and mesodermal differentiation during organoid maturation [42]. The NMOs were used to reproduce the key features of myasthenia gravis pathology, showing promising effects on ALS modeling and treatment development.

Zhao et al. built and compared the cerebral organoids using iPSCs with Apolipoprotein E (APOE) $\epsilon 3/\epsilon 3$ or $\epsilon 4/\epsilon 4$ genotype from both healthy people and AD patients [43]. Increased levels of Amyloid β -protein ($A\beta$) and phosphorylated tau were found in AD patient-derived cerebral organoids, and APOE4 exacerbated tau pathology in both organoid models. Interestingly, APOE4-related phenotypes were attenuated in cerebral organoids from AD patients within the isogenic conversion of APOE4 to APOE3. Although those organoid models provide comprehensive tools for personalized iPSCs-based disease investigations, serum exposure from the blood-brain-barrier (BBB) leakage, a well-known risk factor for AD, cannot be replicated without the bloodstream in organoid systems. Chen et al. reconstructed a cerebral organoid mimicking the BBB leakage by serum exposure and investigated the relevant risk factors for AD [44]. It was found that $A\beta$ (relative level from 1 to 10) and p-Tau levels (relative level from 1 to 2) could be regulated by serum exposure through inducing glycogen synthase kinase-3 α/β (GSK3 α/β) and beta-secretase 1 (BACE) levels, while synaptic function was reduced in both neurons and astrocytes (Fig. 2A). In another work, N. Wulansari used human embryonic stem cells edited by CRISPR-associated protein 9 (Cas9)-mediated gene to establish PD organoids [45]. Typical PD pathologic features were reproduced by introducing the DNAJC6 mutation in human midbrain-like organoids (hMLOs). For example, pathological α -synuclein was aggregated and midbrain-type dopamine neurons were degenerated, increasing the intrinsic neuronal firing frequency (more than twice that of wild-type hMLOs at DIV80) and disabling the mitochondrial and lysosomal (Fig. 2B).

It has to be mentioned that it is still challenging to characterize the neural activity from large organoids due to the lack of suitable devices, despite that the huge neural activities are key index for normal neural

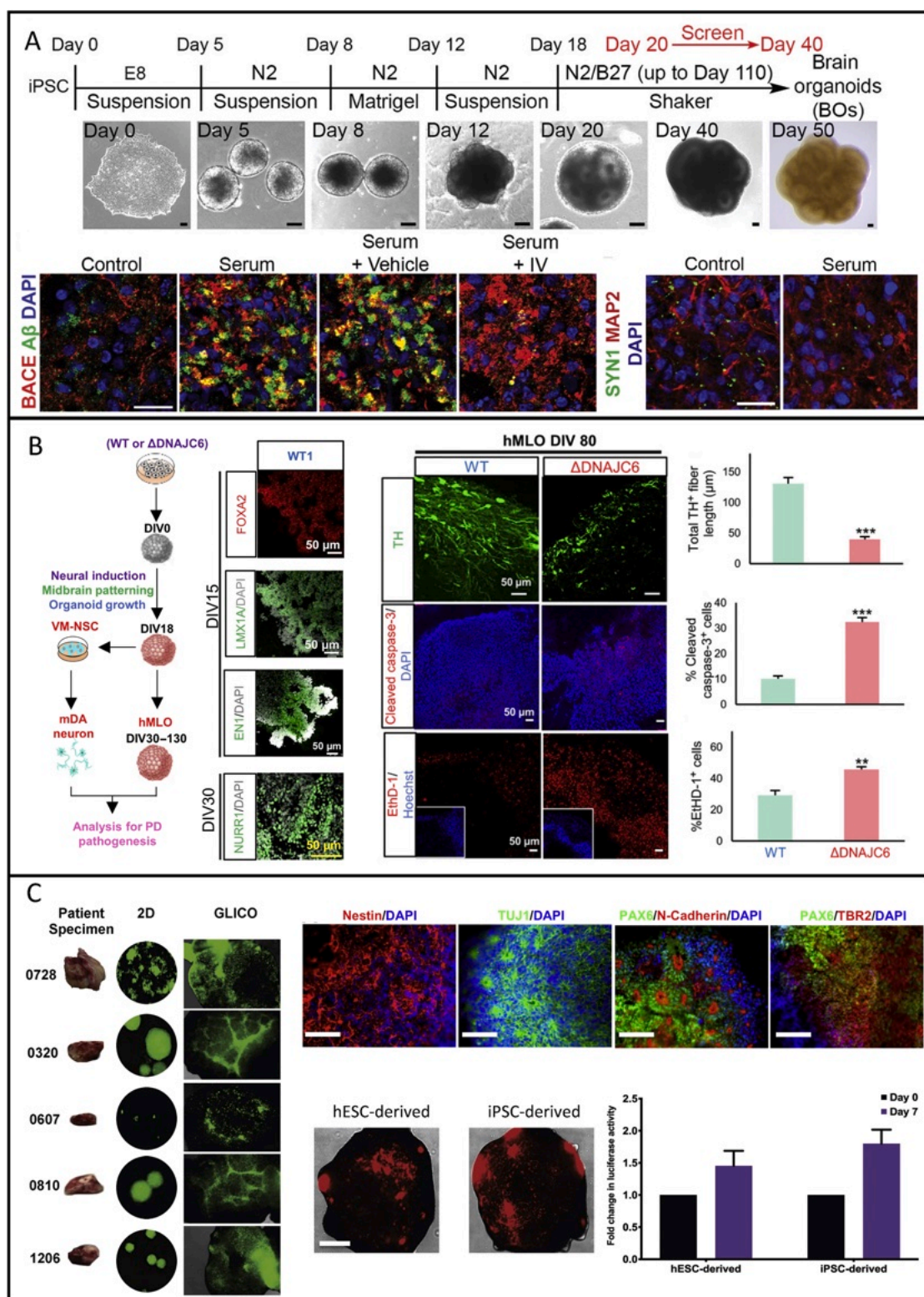


Fig. 2. CNS organoids for neurological diseases and brain tumors. (A) The development of hiPSCs-derived brain organoids at different stages, and serum exposure led to the Aβ-like pathology and synaptic loss. Reproduced from [44] under the terms of the Creative Commons CC BY license. (B) Schematic of isogenic human embryonic stem cells (hESCs) differentiation and dopamine neuron degeneration. Reproduced from [45] under a CC BY-NC 4.0 license, with permission from AAAS. (C) Historical development of cerebral organoid glioma derived from six patient tumor specimens. The related markers at specific neural development were expressed, and GLICO tumor could grow in either 1-month-old hESC-derived (H1) or iPSC-derived (H6) organoids [46]. Copyright 2019, Elsevier.

functions which might be restricted in neurodegenerative patients. Wu et al. designed a novel three-dimensional liquid metal-based neural interface (mMPC) that combines with human hippocampal organoids (hHOs) to non-invasively obtain electrophysiological information from these free-floating organoids [47]. Induced by Wnt3a and SHH activators, hHOs containing many PAX6⁺, HOPX⁺ hippocampal progenitor cells and PROX1⁺ granular neurons were generated. The transcriptome features of these organs are highly correlated with the human hippocampus. Taken together, the human brain organoids provide an unprecedented tool for the investigation of underlying mechanisms and the advancement of therapeutic approaches for neurological diseases.

2.2.4. CNS organoids for brain tumors

Malignant brain tumors, that is brain cancers, are one of the most devastating cancers with the greatest lethality. Organoid tumor models have been developed and are essential to investigate the nature of the neoplastic disease. More importantly, such biomimetic 3D systems can open a novel avenue to evaluate the effectiveness of candidate therapeutic agents. Brain tumor organoids can be classified as genetically-engineered models or models derived from dissected tumors or cancer cells. For example, a neoplastic cerebral organoid (neoCOR) was constructed by introducing oncogenic transposon- and CRISPR–Cas9-mediated mutagenesis [48]. Brain tumorigenesis was replicated by identifying neoplasms resembling glioblastoma and central nervous system primitive neuroectodermal tumors (CNS-PNET), which are clinically relevant mutations. Tumor biology and the effects of drugs on specific DNA aberrations are suitable to be investigated on the above neoCOR, providing a promising basic and preclinical brain tumor model. Henriks et al. established human fetal brain organoids (FeBOs), used CRISPR technology to introduce specific gene mutations in FeBOs, such as the knockout of genes such as TP53, and successfully created a gene mutation model associated with brain tumors, which was able to simulate the growth advantage, invasiveness and other properties of brain tumor cells. It provides a powerful platform for discovering new therapeutic targets, drug screening and exploring the molecular mechanism of brain tumors [49]. In another study, a cerebral organoid glioma (GLICO) model that used hESC introduced from patient-derived glioma stem cells (GSCs) was established [46]. The GSCs invasion, proliferation and tumor formation could be found in the GLICO system, showing a patient-like phenotype (Fig. 2C). Peng et al. cultured patient-derived tumor explants iPSC-derived cerebral organoids, providing a large variety of human tumors that can be cultured in CNS, to better recapitulate the cellular and molecular pathology of human brain tumors [50].

In summary, CNS organoids can provide models of neural tissues that are closer to their natural state, providing an efficient strategy for reproducing the pathological process of neurological diseases *in vitro* and revealing the pathophysiology of the conditions, alongside evaluating the impacts of various pharmaceutical agents on neuronal cells, predicting the effects of drugs in the human body, and accelerating the process of drug development. However, current CNS organoids cannot fully simulate the complex 3D neural network of the CNS and replicate all complex cell interactions and signaling pathways. There is a need to work on improving the complexity, stability, and long-term culture capabilities of CNS organoids to make them closer to the real nervous system. In addition, CNS organoids often lack a blood supply, limiting organoid growth and function. The formation of a vascular network inside the organoids helps provide oxygen, nutrients and metabolic products required by the cells, and should promote the growth and maturation of organoids. In the future, CNS organoids may be personalized according to the patient's specific genotype or disease phenotype, thereby more accurately simulating the patient's disease state and providing a basis for developing personalized treatment plans.

2.3. Engineered physiological and pathological CNS-on-Chip models

Despite the promising advantages of organoids in establishing

physiological and pathological models of CNS *in vitro*, they still have obvious weaknesses such as uncontrollable ECM or relatively smaller size compared with human organs. Particularly, it is challenging to increase the survival rate of large organoids (millimetre-scale) because of the difficulties in nutrient diffusion [51]. Organ-on-chip approaches may provide another path to address some of the limitations of organoids for the reconstruction of CNS. First, a dynamic microfluidic system is integrated and thus fresh media can be continuously perfused into the cultured tissues, which can better mimic the interactions of cells and body fluid and improve cell viability. Second, the design of microfluidic systems can be bioengineered better by incorporating compartmentalized neuronal chambers with multiple types of cells and variable microchannels, which replicate the cross-talks of several tissues as a whole system. Third, microelectrodes can be integrated into the chips optionally to realize in-situ stimulation, sensing, and nondestructive testing for CNS as electrophysiological tissues. Finally, higher throughput drug testing and screening can be achieved on those microfluidic devices.

2.3.1. CNS-on-Chip models for neuronal regeneration

Understanding the internal mechanisms of physiological activities such as neuronal migration, neurite outgrowth, synapse formation and myelination is important for developing the treatment for CNS diseases. Microfluidic organ-on-a-chip (OoC) models represent golden opportunities to study neural physiology with controllable environmental factors, and examples include chemical gradients, as well as extrinsic mechanical or electrical stimuli. For instance, a synapse-on-a-chip model was fabricated with two parallel and lateral compartments constraining the neuronal somas, and microchannels guiding axons and dendrites to build synaptic connections [52]. Once neurons in one lateral compartment were stimulated by a chemical compound, an evident response would be produced by neurons in the other compartment, showing functional synapses have been successfully constructed. Besides, orientated connecting conduit could be mimicked by using the microfluidic technique via designing and building the cell culture chamber and fluid walls in arrangement. Nebuloni et al. designed a chip having dumbbell-shaped circuits and using it successfully mimicked unidirectional cortico-striatal connectivity, which is critical for their functioning but is often oversimplified by *in vitro* co-cultures [53]. The pro-regenerative conditions on axonal regrowth were compared and clarified in that work.

In addition to the simulation of physical and chemical factors, CNS-on-chip techniques are able to replicate CNS functions such as the myelination process *in vitro*, which is usually hindered by the random distribution of neurites and myelin within the conventional cell culture plate. Kerman et al. fabricated a myelin-on-chip device involving embryonic stem cell-derived neurons and myelinating oligodendrocytes in separated open neuronal compartments (Fig. 3A) [54]. In 2 weeks, elongated axons were ensheathed by oligodendrocytes, forming the myelin (percent of myelinated axons: 2.61 %). Furthermore, the manipulation of stem cells and their encapsulation within microfluidic chips have demonstrated more favorable outcomes than traditional treatments in treating spinal cord injuries. Mohammadi et al. used microRNA-7 (miR-7) to transduce trabecular meshwork mesenchymal stem/stromal cells (TMMSCs) and encapsulated them in alginate-reduced graphene oxide (alginate-rGO) hydrogel via a microfluidic chip to explore its therapeutic efficacy in an animal model of spinal cord injury [55]. In contrast to 2D culture, miR-7–3D was found to enhance locomotion capabilities, reduce cavity size, and increase myelination in contusion SCI rats, illustrating that the confluence of heightened expression of miR-7 alongside the entrapment of TMMSCs within hydrogel matrices holds potential as a novel and auspicious approach for treating SCI.

2.3.2. BBB-on-Chip models

The BBB represent almost the most significant barrier in brain

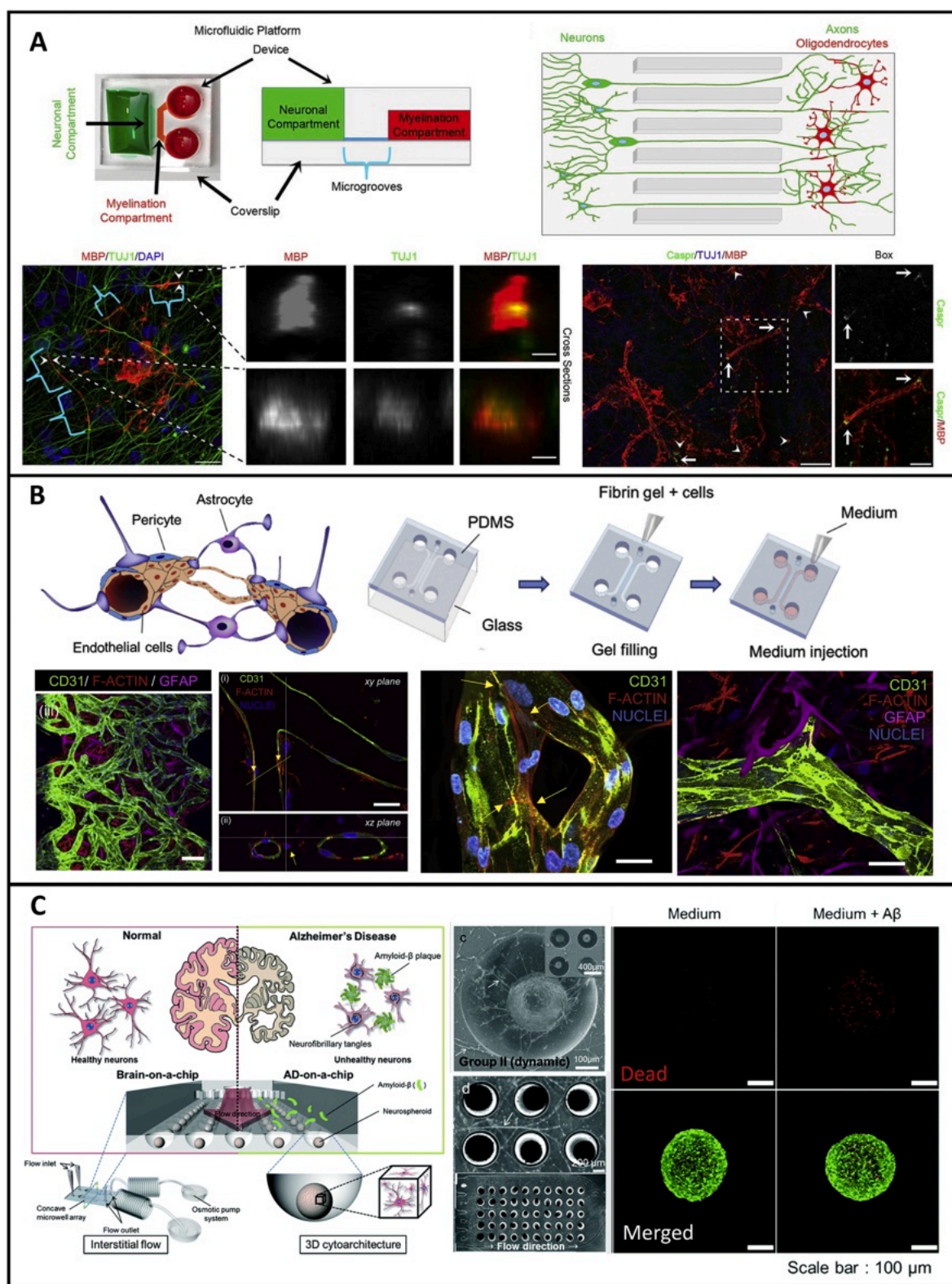


Fig. 3. CNS-on-Chip models for neuronal regeneration, 3D microvascular network and pathological modeling. (A) Neurons and oligodendrocytes were respectively plated into the chip compartments and axons can extend along the microgrooves to form myelin sheath in the myelination compartment. Related key markers were stained [54]. Copyright 2015, Company Biologists. (B) Schematic representation of the blood-brain barrier (BBB) chip and 3D microvascular network (μVN). Contact interactions of pericytes and enveloped brain microvessels were produced. Reproduced from [56] under the terms of the Creative Commons CC BY license. (C) A three-dimensional brain-on-a-chip was fabricated with the interstitial level of flow. The neural network was formed and cell viability of neurospheroids with and without amyloid-β was tested [57]. Copyright 2015, Royal Society of Chemistry.

disease therapy because therapeutic molecules can hardly pass through it to reach the targeted cerebrum. Within an intricate configuration, BBB serves as a regulatory barrier, demarcating the central nervous system from systemic circulation. BBB is made up of diverse cellular constituents encompassing glial cells, endothelial cells, and pericytes. It is obvious that BBB dysfunction will result in turbulence in the CNS environment and subsequent neurological disorders [58]. As one of the complex neurovascular units, BBB-on-chip models are usually constructed to mimic the structural junctions and functional interplays between neuronal cells and vascular endothelial cells. The design of BBB-on-chip systems includes parallel, vertical and full-contact. Refaat et al. used highly sensitive radioisotope assays to assess whether microfluidic blood-brain barrier chips (μ BBB) could accurately predict the brain permeability of drugs. By comparing the B/P ratios of the nine radiolabeled compounds *in vivo* with the μ BBB model, the results showed a strong positive correlation between the μ BBB model and the *in vivo* outcomes, indicating that the μ BBB model has a high degree of predictive power and is able to predict the ability of the drug to penetrate the brain [59].

Vertical design consists of upper and lower chambers culturing vascular and neural cells, respectively, which are partitioned by a porous membrane. For example, Ahn et al. designed a vertical BBB microphysiological platform that rebuilt the cellular interactions, low permeability and key gene expressions. Reactive gliosis was reduced and aquaporin-4 (AQP4) distribution was polarized, forming a 3D astrocytic network [60]. Compared with human astrocytes (HAs) cultured on 2D matrigel-coated surfaces, HAs cultivated within a three-dimensional matrigel matrix displayed a morphology reminiscent of *in vivo* conditions, characterized by a more extensively branched structure, with smaller cell bodies and radial distribution of branches. With the capability of 3D mapping of nanoparticle distributions, this platform could demonstrate the BBB penetrations through transcytosis mediated by receptors accurately.

In the adjacent conduits of the parallel configuration, an array of micropillars or microchannels is usually employed to separate two chambers. Lyuwe et al. reported a BBB-on-chip device mimicking a functional neurovascular unit for ischemic stroke investigation with three parallel channels, namely the blood-side, brain and cerebrospinal fluid (CSF)-side channels [61]. The authors characterized the expression levels of genes related to post-stroke pathologies. The ingress of selected candidate stem cell populations and the transcriptional activity of genes associated with post-stroke pathophysiology was fully characterized. It was found that the integrity of the neurovascular unit structure and function instead of the neuron regeneration had a significant impact on the recovery of synaptic activities. Li et al. generated a 3D microchannel-BBB device which could replicate the response of rat cerebral or mesenteric microvessels to plasma proteins, cytokines, VEGF, and the enzyme heparinase III in terms of permeability and glycoalyx (heparan sulfate) [62].

Those vertical and parallel designs contain microchannels with rectangular cross-sections, which may cause an inhomogeneous distribution of the shear stress on the vascular endothelium. Omnidirectional contact of different types of cells by using full-contact design of microfluidic platforms could hence be developed. For example, 3D cerebral vascular networks were reproduced by direct neurovascular generation from co-culture and self-assembly of pericytes and/or astrocytes in hydrogel (Fig. 3B) [56].

2.3.3. CNS-on-Chip systems for pathological modeling

Neural network disruption and pathologies result from many factors such as CNS cellular microenvironment alterations, cell death, ageing, etc. In order to discern the fundamental mechanisms governing central nervous system pathologies, it is imperative to replicate closely the disease-related processes *in vitro*. OoC systems using hiPSCs from patients offer human-specific mechanistic insights into the treatment development for CNS disorders. The study on OoC models of traumatic

injury includes axonal injury and astrocyte injury. Inflammation causes microglia accumulation during the regeneration of brain or spinal cord tissues. In one study, rat embryonic hippocampal neurons were cultured on a microfluidic device and it was found that mechanical axotomy could downregulate Netrin-1 and subsequently lose hallmarks of injury *in vivo* [63]. In another study, the consequences of simulated blast-induced traumatic brain injury on astrocytes were investigated by employing a regulated electrical discharge apparatus [64].

In addition to traumatic injury, neurodegenerative diseases such as AD, motor neuron disease and PD are threatening human health conditions with a growing trend for which there is still no effective treatment. Although organoid models have been built for the investigation of neurodegenerative diseases, they lack interstitial flow which is also known to affect cell-cell communication between non-synaptic neurons in the brain. Therefore, many researchers have built up biomimetic CNS-on-chip models which serve as potential tools to accelerate the exploration of personalized therapeutics. AD is a degenerative disease caused by the aggregation of amyloid peptides and the phosphorylation of Tau microtubule proteins within the cerebral tissue. An AD chip was developed with a continuous fluidic flux designed to replicate the interstitial milieu of the brain [57]. Neurospheroids were cultured in concave microwell arrays, and their size, differentiation and network formation influenced by flow were investigated. The toxic effects of amyloid- β were tested and analyzed because this molecule has been identified to be the major cause of AD disease (Fig. 3C). PD is generally considered to be induced by the dopaminergic neuron loss in substantia nigra, resulting in motor disturbances encompassing both tremor and bradykinesia. In another study, PD patient-derived human neuroepithelial stem cells were cultured on a PD-on-a-chip to observe their differentiation to dopaminergic neurons by Ca^{2+} imaging and immunostaining assay [65]. The software was integrated with the system for analyzing patient-specific models within different perturbations over the long term, which is of great significance for personalized disease modeling. Huntington's disease (HD) is expressed by the dysfunction of the corticostriatal network and is considered to be a devastating neurodegenerative disorder. Virlogeux et al. [66] built up a microfluidic device with subcellular resolution to observe the defect conditions in each compartment of the corticostriatal circuit, and it was found that the circuit could be altered or restored by the genetic profile of the pre-synaptic domain to a great extent.

Overall, CNS-on-chip devices are capable of precisely controlling the growth environment of neural cells as well as intercellular connections and communication, and this highly bionic environment helps to more accurately study the functions, disease mechanisms and regenerative processes of the nervous system. In addition, CNS-on-chip models provide a highly controllable setting that can apply drugs to neural cells in the chip at high throughput and offer the capability to monitor the effects of drugs on cells and neural networks, thus rapidly assessing drug efficacy and toxicity and improving the efficiency of drug development. Similar to CNS organoids, CNS-on-chip models based on microfluidic chip technology are not able to fully reproduce the complex neural network structure built by a large number of neurons and multiple cell types in the CNS. CNS-on-chip systems can be combined with biomimetic complex structures composed of multiple types of nerve cells and scaffold materials constructed through bioprinting technology to further mimic the complex structure of the CNS. Future developments could also allow the construction of modular microfluidic chips containing the blood supply system, immune system, and other organs to more accurately simulate the effects of the whole body's physiological environment on the CNS and to enhance the reliability of CNS-on-chip devices.

2.4. 3D printed Scaffold-based physiological and pathological models of CNS using 3D printing

One of the foremost hurdles in the advancement of neural model

development is to reconstruct or capture the structural and cellular characteristics of the complex human nervous system accurately *in vitro* [67]. 3D printing can accurately arrange multiple types of cells, reconstruct complex CNS structures, and simulate the spatial organization and interaction of cells in the central nervous system. It can also provide extra benefits for building neural models by presenting topographic cues, which promote neurocyte elongation and migration along the desired direction. Bioprinting with bioinks encapsulating neural cells can lead to precise cell arrangements for physiologically mimicking CNS and its functions [68]. Bioinks are essential to influence the mechanical properties, biocompatibility and degradation characteristics of 3D-printed scaffolds [69]. Numerous natural polymers have been successfully employed for 3D printed neural models due to their degradability by natural enzymes and superior biocompatibility such as alginate [70], chitosan [71], collagen [72], gelatin [73], and hyaluronic acid [74]. Compared with natural biomaterials, synthetic polymers such as poly(glycerol sebacate) [75], poly-ε-caprolactone (PCL) [76], polyurethane (PU) [77] and poly-lactic-co-glycolic acid (PLGA) [78] provide tailorable chemical properties and optimized mechanical performance, hydrophilicity and degradation rate via copolymerization, which makes them attractive for 3D printing of scaffold-based neural models.

Advanced research efforts have proposed a variety of CNS models such as bioprinted living constructs and organoids, and 3D-printed culture platforms, resulting in suitable performance in both morphological and functional mimics. For example, Joung et al. bioprinted a spinal cord-mimic construct with iPSC-derived spinal neuronal progenitor cells (sNPCs) and oligodendrocyte progenitor cells (OPCs), and those cells were precisely distributed in different positions during the printing process [79]. Intracellular calcium influxes can be integrated after 2 weeks of printing by Fluo-4 calcium imaging. Furthermore, brain-like structures with multiple layers were constructed to mimic the different brain regions using primary cortical neurons- containing bioink. Primary cortical neurons and glial cells were successfully encapsulated, and they survived and formed a network. In addition, 3D printing technology can be used to construct bionic-engineered spinal cord structures for spinal cord injury repair. Liu and colleagues engineered an innovative bioink formulation comprising active chitosan, derivatives of hyaluronic acid, and a matrix exhibiting prompt gelation (occurring within 20 s) alongside inherent covalent cross-linking properties to precisely print living neural tissues with intricate configurations and spatial arrangements of neural progenitor cells (NSCs) which maintained high viability (~95 %) [80]. The spinal cord-like construct promoted axonal regeneration, reduced glial scar deposition, and significantly restored locomotor ability in a SCI rat model. In reported research, Wang et al. placed bone marrow mesenchymal stem cells (BMSCs) and rat Schwann cells (RSCs) in specific spatial locations to form a bionic spinal cord to promote neuronal connectivity and neural differentiation by 3D bioprinting [81]. In this bionic spinal cord scaffold, BMSCs differentiated into neurons with axonal projections extending through the scaffold channels while RSCs differentiated into oligodendrocytes with myelinated axons, with the capability to rebuild spinal cord conduction pathways at the site of spinal cord injury and capable of promoting the recovery of motor function. However, the mechanical properties of bioinks need to mimic the super soft native CNS tissue to support cell growth and function, but often lack mechanical stability, leading to the collapse or deformation of printed 3D structures. Therefore, it is challenging to balance the softness with the stability of bioinks and difficult to construct complex 3D architectures.

A recent study used lipid-bilayer-supported 3D printing technology to construct human neural stem cells (hNSCs) and human astrocytes co-culture systems with precise pre patterning (Fig. 4A) [82]. It demonstrated the outgrowth and migration process of hNSCs into the cell-free matrix and into the astrocyte-containing matrix, which could provide an effective model to investigate human cerebral cortex development. Similarly, Janzen et al. printed PCL microfiber frames with ultra-soft matrices encapsulated with neurons as an innovative

three-dimensional model for investigating the pathophysiology of cerebral disorders [83]. It was found that astrocytes seeded in the matrix could improve neuronal viability and better reproduce the brain-like structure. Those composite structures would provide additional mechanical support for printing soft CNS tissues in 3D

Regarding the manufacture of organ-on-chip systems, 3D printing facilitates the fabrication process of fine structures and sophisticated designs. In one study, stereolithography (SL) technology was adopted to print a brain-on-chip device in the micrometer range enabling dynamic perfusion [84]. The development of a neural-vascular network was observed and investigated by culturing cerebral organoids on the platform, suggesting a potential system for organoid vascularization and drug testing for BBB (Fig. 4B). Endothelial cells invaded the edges of brain organoids in 83 % of co-cultivated organoids, infiltration was observed, while penetration into the core of the organoid occurred in 42 % of instances. In similar research, Cai et al. printed an organoid holder device and integrated it into a spinal cord-on-a-chip model [85]. In this way, the microphysiological systems were compatible with multiwell-plates and multi-electrode array (MEA) systems for the facilitation of seamless neural activity quantification (Fig. 4C). The planarized organoids exhibited increased electrode engagement compared to conventional organoids, thereby enhancing the interface between the organoid and electrodes for improved neural activity assessments and reducing necrosis and hypoxia. The average firing frequency exhibited by these planarized organoids (0.0658 ± 0.0234 Hz) surpassed that of conventional organoids by nearly tenfold (0.0068 ± 0.0071 Hz, $p = 0.00082$), which may be attributed to the increased neural activity in human spinal microphysiological systems because of enhanced perfusion provided by the flattened organoids design.

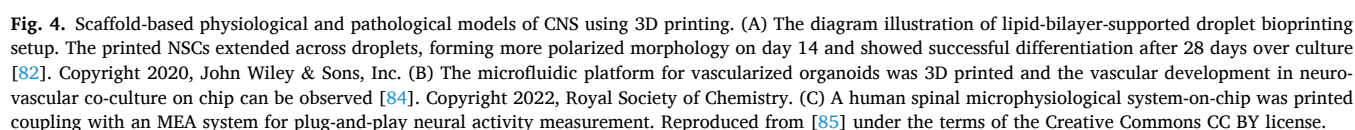
Although 3D printing can reconstruct certain features of the central nervous system, its complexity is still limited, and simulating the intricate configuration and functionality of the entirety of the central nervous system continues to pose a challenge. During the 3D printing process, cell survival and functional maintenance may be affected, and maintaining the long-term stability and vitality of cells remains a problem. Furthermore, current central nervous system models may not fully reproduce the natural central nervous system complexity, such as the lack of blood supply and the integrity of other surrounding tissues. As technology continues to advance, it can be anticipated that 3D printing will become more precise and higher-resolution of printed structures will be achieved, able to better reconstruct the microstructure and function of the natural CNS. In the future, 3D printing can be combined with other technologies, such as biosensors, optical imaging and microfluidic technology, to further improve the functionality and application range of CNS models.

In conclusion, advances in CNS *in vitro* models have enabled precise replication of neural complexity, leveraging iPSC-derived organoids to mimic developmental processes (e.g., cortical layering, spinal cord dorsoventral patterning) and pathologies like AD, PD, ALS and brain tumors. CNS-on-chip systems integrate microfluidics to model dynamic microenvironments, such as BBB permeability and synaptic connectivity, with embedded sensors monitoring drug responses and neural activity in real time. 3D-printed scaffolds with spatially organized neural cells promote axonal regeneration and myelination in spinal cord injury models, restoring locomotor function. These innovations bridge *in vitro* models with human pathophysiology, accelerating drug discovery and personalized therapies for neurodegenerative, psychiatric, and traumatic CNS disorders.

3. PNS modeling for neural regeneration and disease treatment

3.1. Requirements for *in vitro* PNS models

The PNS is a pervasive network, encompassing afferent sensory fibers, efferent motor fibers, as well as interneurons that innervates the entire body. Functions such as motor control, information transmission,



immune modulation, and energy conservation are all dominated by the PNS [86]. Neurons are key to the establishment of a PNS model. Sensory neurons are responsible for transmitting peripheral sensory information to the CNS, while motor neurons transmit central commands to effectors to achieve sensory conduction and motor control simulation. SCs, as glial cells unique to the peripheral nervous system, can not only form myelin to insulate and protect nerve fibers, but also secrete neurotrophic factors to promote nerve regeneration. The two types of cells work

together to ensure that the constructed model accurately simulates the physiological functions and injury repair processes of the peripheral nervous system. In addition, there are certain requirements for the cell structure of neurons and SCs when the PNS model is established. Neurons need to fully present dendrites, axons, and cell bodies. Dendrites affect signal reception, axon structural characteristics determine the speed of nerve impulse conduction, and the cell body is a metabolic center. The myelin sheath is formed by SCs, and its thickness is related to

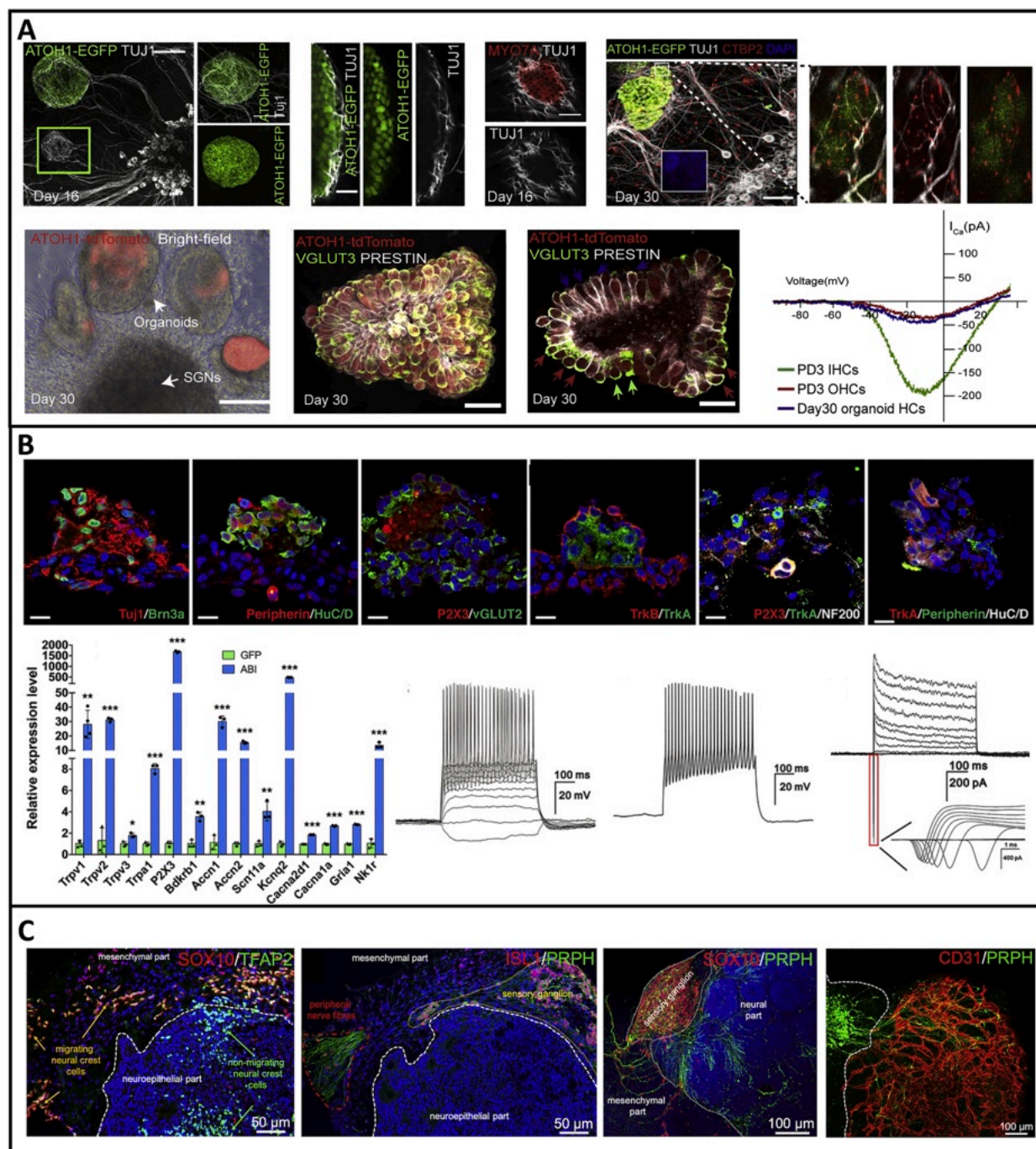


Fig. 5. Microphysiological and pathological organoids of PNS. (A) SGN-innervated cochlear organoids were generated and functional auditory circuits were formed. The morphological and electrophysiological properties were recomplified in HCs organoids [87]. Copyright 2023, Elsevier. (B) Mostly peripheral sensory neurons can be induced in iSGs by ABI, showing the functional properties of sensory neurons. Reproduced from [88] under a CC BY-NC 4.0 license, with permission from AAAS. (C) The peripheral nerve and sympathetic ganglia were formed in neuro-mesodermal assembloids, and complex vascular plexus can be observed [90]. Copyright 2023, Elsevier.

the speed of nerve signal conduction. The segmental structure of the Ranvier nodes in the myelin sheath gap is of great significance to the jumping conduction of nerve impulses. The precise construction of the two structures can effectively simulate the signal conduction function of PNS.

When the PNS model is established, in terms of mechanical properties, cells and extracellular matrix are required to have appropriate elasticity and toughness to simulate the stretching, bending deformation and resistance to external impact of nerves in physiological activities. In terms of biochemical properties, it is necessary to ensure that neurons can normally synthesize and release neurotransmitters to achieve signal transmission, and that cells have the ability to secrete and respond to neurotrophic factors to simulate the process of nerve development and repair. At the same time, the ion channel function is accurately simulated to present the electrophysiological activity of the nerves. The mechanical and biochemical properties work together to achieve effective simulation of the function of the peripheral nervous system.

3.2. Microphysiological physiological and pathological organoids of PNS

Organoids offer superior advantages in reproducing the PNS structure by involving multiple cell types and reproducing physiologically relevant 3D environments, not only facilitating the understanding and deciphering of the mechanisms behind the functional maturation of relevant tissues but also establishing peripheral disease models for understanding the pathogenesis of diseases and for testing potential treatments. For example, recent research has focused on the development of cochlear organoids aiming to replicate peripheral auditory circuitry functions within a 3D co-culture system [87]. A stepwise induction process was employed to guide the differentiation of cochlear hair cells (HCs), while simultaneously promoting the growth of neurites from spiral ganglion neurons (SGNs) to form functional synapses (Fig. 5A). After 30 days' culture, Ca^{2+} currents of organoid HCs could reach about -60 mV, with a maximum value of -10 mV after activation, similar to that of outer hair cells. In another study, self-organized sensory ganglion organoids were created by combining triple *Brn3b/3a*, TFs *Ascl1* and *Isl1* (ABI) [88]. These organoids exhibited key features of peripheral sensory neurons such as subtype diversity, gene expression, electrophysiological functions, and innervation patterns (Fig. 5B). The immunostaining results showed that the organoids were co-labeled with typical proteins of sensory neuron types such as $\text{P2} \times 3+$ neurons and $\text{vGLUT2}+$ neurons, and the corresponding receptor ion channel genes presented up-regulated expression. Furthermore, electrophysiological recordings were performed to capture the responses of neurons both inside and outside the induced sensory ganglion (iSG). Typical sodium and potassium currents were tested on the majority of neurons, and action potential responses were still exhibited after 2 weeks. Lu et al. successfully established a functional human dorsal root ganglion (DRG) organoid by simulating programmed activation of signaling pathways *in vivo*, which was able to simulate human DRG development and generate multiple sensory neuron types and glial cells. It provides a powerful tool for studying the development of the human sensory nervous system, disease modeling and drug screening [89]. Although the above organoids closely mimic the human PNS neurogenesis process, further specialization and maturation of sensory neurons could be limited by the lack of complex neural networks. To enhance the fidelity of the tissue complexity replication and facilitate inter-tissue interactions, it is possible to construct organoid assemblies by combining multiple organoids together. Rockel et al. developed a neuro-mesodermal assembloid model to reproduce the PNS development process [90]. Fig. 5C shows bundles of peripheral nerve fibers projected into the mesenchymal tissue from the sensory ganglia, accompanied by $\text{SOX10}+$ cells. Interestingly, the assembloid showed a closed interaction of sensory ganglia and dense $\text{CD31}+$ vascular network, which was formed through the co-development of migrating NCCs and embryonically derived peripheral vascular plexus [91].

Overall, PNS organoids can reconstruct the structure of peripheral nerve tissue of multiple cell types in a 3D environment and simulate the functional maturation process under physiological conditions, providing an important tool for studying peripheral nerve development processes and disease mechanisms. In the future, a broader range of cell types should be integrated into the construction of PNS organoids to enhance the accuracy of mimicking the intricate multicellular interactions and complex architectures of the PNS. Meanwhile, more effective methods should be developed to obtain large amounts and high-quality cells, and the development level of cell engineering technology should be further improved.

3.3. Engineered physiological and pathological PNS-on-Chip models

Peripheral nerve tissue-on-a-chip platforms have been constructed by various microfabrication methods combining microfluidic techniques. These dynamic platforms are able to capture organism-level responses by mimicking the nutrient exchange and waste metabolism during the interactions of the peripheral system and the other organs it dominates [92].

3.3.1. PNS-On-Chip systems for neural regeneration

Neurite outgrowth models are of great value in deepening our comprehension of PNS regeneration processes and finding effective therapies for treating nerve injuries. For instance, a microdevice was developed, comprising a specialized chamber for cultivating spheroids, a microchannel that facilitates the elongation of axons, and a target chamber designed to receive the axon terminals [93]. After 20 days of culture, the fascicle structure was formed with single, straight, and unidirectional morphology. Axon fascicle formation achieved a remarkable efficiency of over 90 % by the 30th day in three different lines of hiPSCs, derived from individuals with varying backgrounds. In another study, a 3D motor neuron model that allows spatially segregated interrogation of axons and motor neuron somata was developed to precisely quantify the axonal outgrowth and thus test its response to chemotherapeutic drugs such as vincristine [94]. To further promote axonal growth in an orientated direction beyond the microgrooves for better quantitative analysis of axonal regeneration, Lee et al. designed a peripheral nerve injury-on-a-chip having aligned electrospun with a microfluidic device. The platform could guide dorsal root ganglion's axonal regeneration and monitor multiple axons after drug treatment [95]. To build a more systematic preclinical evaluation platform to study the PNS repair performance, Yu et al. designed a nerve-on-chip platform with a multitrack architecture, incorporating core-shell microfibers that possess adjustable functionalities [96]. As a result, it was possible to screen out microfibers that were capable of improving nerve regeneration appropriately and effectively. Moreover, the optimized microfibers showed a rapid promotion ability for *in vivo* nerve regeneration and function recovery (Fig. 6A). Despite those advances in PNS-on-chip models for stimulating neural development and regeneration, it is still difficult to model diverse PNS neuron subtypes (e.g., sensory vs. motor neurons) and their distinct regeneration pathways is technically demanding. Besides, crosstalk with other types of cells needed to be considered to rebuild the inflammatory and trophic signaling.

3.3.2. PNS-On-chip platforms for PNS myelination

Myelin represents a distinctive type of plasma membrane, crafted either by oligodendrocytes in the CNS or Schwann cells (SCs) in the PNS, which covers axons and enhances nerve conduction 20–100-fold faster compared to that in the unmyelinated axons [98,99]. Besides, myelin also plays a role in regulating the maturation and electrical properties by providing extrinsic trophic signals [100]. Demyelinating diseases, resulting from genetic mutations, inflammation, or exposure to chemical and environmental factors, have profound implications for human well-being. These conditions are characterized by symptoms such as muscle weakness, diminished or absent spinal reflexes, and disruptions

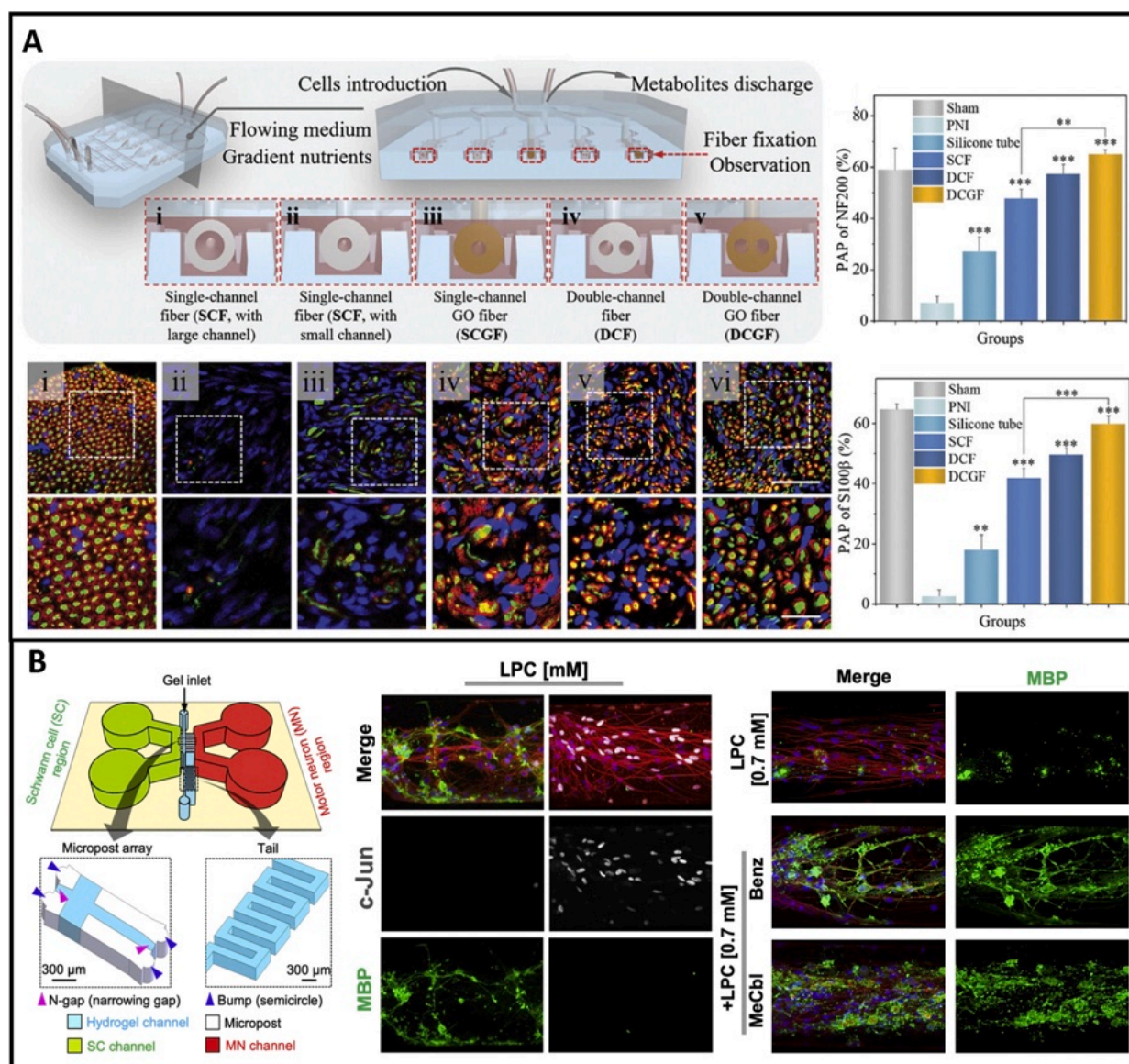


Fig. 6. PNS-On-Chip systems for neural regeneration and PNS myelination. (A) The novel nerve-on-a-chips with fibers fixed inside, and successful axonal regeneration and remyelination were verified via immunofluorescence staining. Reproduced from [96] under the terms of the Creative Commons CC BY license. (B) A 3D PNS disease microfluidic platform was constructed. Dedifferentiation of SCs and the demyelination process can be induced by lysophosphatidylcholine, by showing opposite expression trends of c-Jun⁺ and MB. Rproduced from [97] under a CC BY-NC 4.0 license, with permission from AAAS.

in electrophysiological conduction [101]. A few studies have presented nerve-on-chip models that mimic the loss and regeneration processes of PNS myelin on a single platform. HYUNG et al. [97] constructed a 3D PNS microfluidic platform composed of one hydrogel channel, and separated cell-loading channels for SCs and motor neurons (MNs), respectively (Fig. 6B). Demyelination and remyelination were successfully reproduced by the varied expression of c-Jun (a negative regulator of the myelination) and MBP (expressed in myelinating glial cells). Following treatment with lysophosphatidylcholine (LPC) at a concentration of 0.7 mM, there was a significant reduction in the expression level of MBP. At 20 and 23 days *in vitro*, the MBP expression level was 2.5- and 5.6-fold lower, respectively, compared to the control group. These findings indicate that LPC induces the process of dedifferentiation and demyelination in SCs. To confirm the remyelination of demyelinated SC-MN cocultures, the thickness of nerve fibers was also assessed and analyzed. The platform allowed hydrogel patterning repeatably and co-culture of MNs and SCs stably over 40 days *in vitro*, showing a

promising way to elucidate the myelination mechanisms and its possible biological pathways. To better recapitulate physiologically relevant models, Park et al. presented a PNS chip by embedding a micro-engineered extracellular matrix hydrogel scaffold for co-culture and interaction of MNs and SCs in 3D [102]. In comparison to MNs monoculture, MN-SC co-culture exhibited a higher relative axonal growth rate. On the first day of MN-SC co-culture, the average neurite length reached $145 \pm 16 \mu\text{m}$, which was approximately 3 times longer than that observed in MN monoculture. Controllable myelination and demyelination were realized by treating the model with ascorbic acid and glial growth factor.

In addition, understanding the complex communication between different cell interactions during myelination is fundamental in neuroscience research. Therefore, it is required to develop tools for selectively analyzing and/or probing specific cells and cell portions. To this aim, a multi-compartment microfluidic device was fabricated and evaluated to observe the long-term neuron-like and myelinating maturity from SH-

SY5Y and primary rat Schwann cells [103]. SCs showed slender, elongated morphology and expressed the p75 neurotrophin receptor, which is a neurotrophin receptor involved in neurite outgrowth (average weekly increase of ~ 1 mm) and myelination [104]. Further studies are needed for the co-culture of more physiologically relevant cells (*i.e.* neurons, SCs, muscles) in distinct compartments to investigate physiopathological mechanisms behind neurodegenerative diseases. In particular, electrical and optogenetic stimulations have been coupled in some microfluidic platforms to clarify the effects of myelination on neuronal activity [105]. Apart from those above technique advances, all-human nerve models using human-organoid cells are desired for providing more valuable histological data for human disease modeling and drug discovery. Sharma et al. reported an all-human peripheral nerve-on-a-chip system demonstrating long-term culture and robust neurite outgrowth and effective myelination induced [106].

3.3.3. PNS-On-Chip models for neuromuscular junction

MNs and muscle fiber innervation via neuromuscular junctions (NMJs) contribute to motor functions in our daily lives. Fabrication of *in vitro* 3D systems that mimic NMJ allows for investigating effective treatments for NMJ-related degenerative diseases. Park et al. [107] developed a compartmentalized NMJ microfluidic device using poly (dimethylsiloxane) (PDMS), comprising separated compartments accommodating neuron soma (mouse embryonic stem cell (mESC)-derived motor neurons) and muscle cells (C2C12 myotubes). 100 parallel microchannels were fabricated to build connections between the compartments. The isolated *in vivo* environment was reproduced by the above design, and the successful NMJ formation between these two segregated cells was realized as well. Subsequently, Uzel et al. introduced a microfluidic platform containing three flanked gel regions as microfluidic sections and two medium channels [108]. Myoblast-derived muscle strips and mESCs were cultured in a 3D hydrogel to replicate the physical separation between the spinal cord and limbs. After seeding for 1 day, the outgrowth of motor neurites from the neurospheres could be detected, as indicated by the presence of Hb9GFP+ processes within the gel. However, in the aforementioned model, only around half of the muscle constructs could be effectively stimulated through neuronal excitation. This lower responsiveness may be attributed to the limited accessibility of the myofibers to the approaching growth cone. After seeding for 4 days, the motor neurite outgrowth from the neurospheres was quantified to have a rate of $10.4 \pm 5.6 \mu\text{m h}^{-1}$. Similarly, another NMJs microfluidic platform was constructed with two culture-divided chambers connected by a 1-mm gap [109]. The two chambers allowed for the independent placement of individual neurospheres, and after 24 h, axonal growth was observed. Remarkably, within 5 to 7 days, the axons extended to cover the distance between the neurospheres and the muscle tissue. Innervation of the muscle microtissues as well as muscle striations could be observed by day 10 of co-culture. Furthermore, the dysfunction of neuromuscular transmission was induced to mimic the pathological changes in NMJ function on the chip. This result was achieved by exposing the system to serum from patients with myasthenia gravis (MG). Differential effects from different patients were detected, demonstrating the translational utility of the system. The 3D chips of NMJs provide thus an appropriate approach for investigating the regulatory mechanisms of MN regeneration capacity [110,111].

Compared with PNS organoids, PNS-on-chip platforms can provide peripheral nerves with a physiological environment closer to the *in vivo* environment, and offer a platform for the study of multi-organ interaction mechanisms, which is helpful for fundamental research and for drug screening of peripheral nervous system-related disease models. More PNS-on-chip models, such as pain-on-chip [112], were developed as diagnostic tools for both preclinical and clinical applications. Besides, those systems can be combined with electronic sensors and stimulation devices to enable high-resolution interactions with neural tissue facilitating more accurate research and treatment. In addition, machine

learning and computational models can be combined to integrate large amounts of experimental data to more accurately predict drug effects and disease mechanisms [113].

3.4. 3D-printed Scaffold-based physiological and pathological PNS models using 3D printing

3D-printed PNS models show better morphological and functional behavior, such as nerve signal transmission, cell activity, and tissue development. In one study, a multiscale composite scaffold was fabricated by bioprinting using GC-MS+ nerve growth factor (NGF) as a cell carrier. This approach successfully promoted neurite extension, with approximately $40 \pm 14\%$ of PC12 cell axons reaching an average length of $14.51 \pm 6.86 \mu\text{m}$ [114]. The hydrogel shell, designed to emulate the epineurium layer, protected the SCs responsible for NGF secretion and nerve cell organization. However, immortalized cell lines adopted in that study are less physiologically representative, furthermore, the cells were randomly distributed in the bioinks without replicating the orientated growth of nerve fibers in human tissues. In another study, Malheiro et al. built an *in vitro* model composed of an electrospun scaffold seeded with rat primary SCs to create bands of Büngner (BoB) and primed PC12 cells as neuronal cell population, and 3D cell growth was realized by a fibrin hydrogel embedded with neuronal cells sitting on top of the scaffold (Fig. 7A) [115]. In short-term culture, compared to the 2D scaffold, the axonal density of PC12 cells increased threefold in the 3D printed scaffold. Myelination was achieved after long-term culture, indicating the maturation of neural networks on the models. In long-term culture, compared to the 2D scaffold, the neural axons demonstrated an elevated level of alignment and approximately twice the normalized axonal area within the 3D fibrin scaffold. Furthermore, the models were applied to examine the prevention effect of epalrestat ($1 \mu\text{M}$) on myelin deformation under hyperglycemic conditions. Despite those advances, the above studies failed to realize the PNS functional verification due to the lack of use of functional cells or organoids (*e.g.* iPSCs-derived cells with electrical activity, or sensitivity). A recent study adopted functional nociceptors that exhibit electrical activity and sensitivity to noxious stimuli for building a 3D-scaffold platform. By incorporating endometrial organoids and pancreatic pseudo islets into such a nerve platform, it was possible to investigate the potential to generate nociceptor innervation models (Fig. 7B) [116].

Concluding, 3D scaffolds can be combined with organoids and organ-on-chip technology to provide more precise cell positioning and peripheral nerve tissue construction capabilities, providing a powerful tool for complex disease model construction and fundamental research.

Overall, Advances in PNS *in vitro* models include iPSC-derived organoids replicating physiologically peripheral nerve-relevant 3D environments for disease modeling, and microfluidic PNS-on-chip platforms enable precise neurite outgrowth modeling under dynamic stimulation for rebuilding physiological/pathological processes. 3D-printed scaffolds with biofunctional materials or micro-structures promote aligned axonal growth and myelination. Despite progress, limitations include incomplete subtype diversity (sensory vs. motor neurons) and physiological fidelity in inflammatory signaling. More efforts are needed to be put to refine multicellular interactions and accelerate personalized therapies for neuropathies, trauma, and degenerative PNS disorders.

4. Applications of 3D neural models for drug development

4.1. Basic disease 3D modeling for drug discovery

3D models open a door for establishing effective disease models that would deepen the understanding of fundamental human pathophysiology and the underlying mechanisms. Sourcing cells from patients can be directly applied to those *in vitro* models to identify individual differences in disease for better medical guidance. Currently, drug

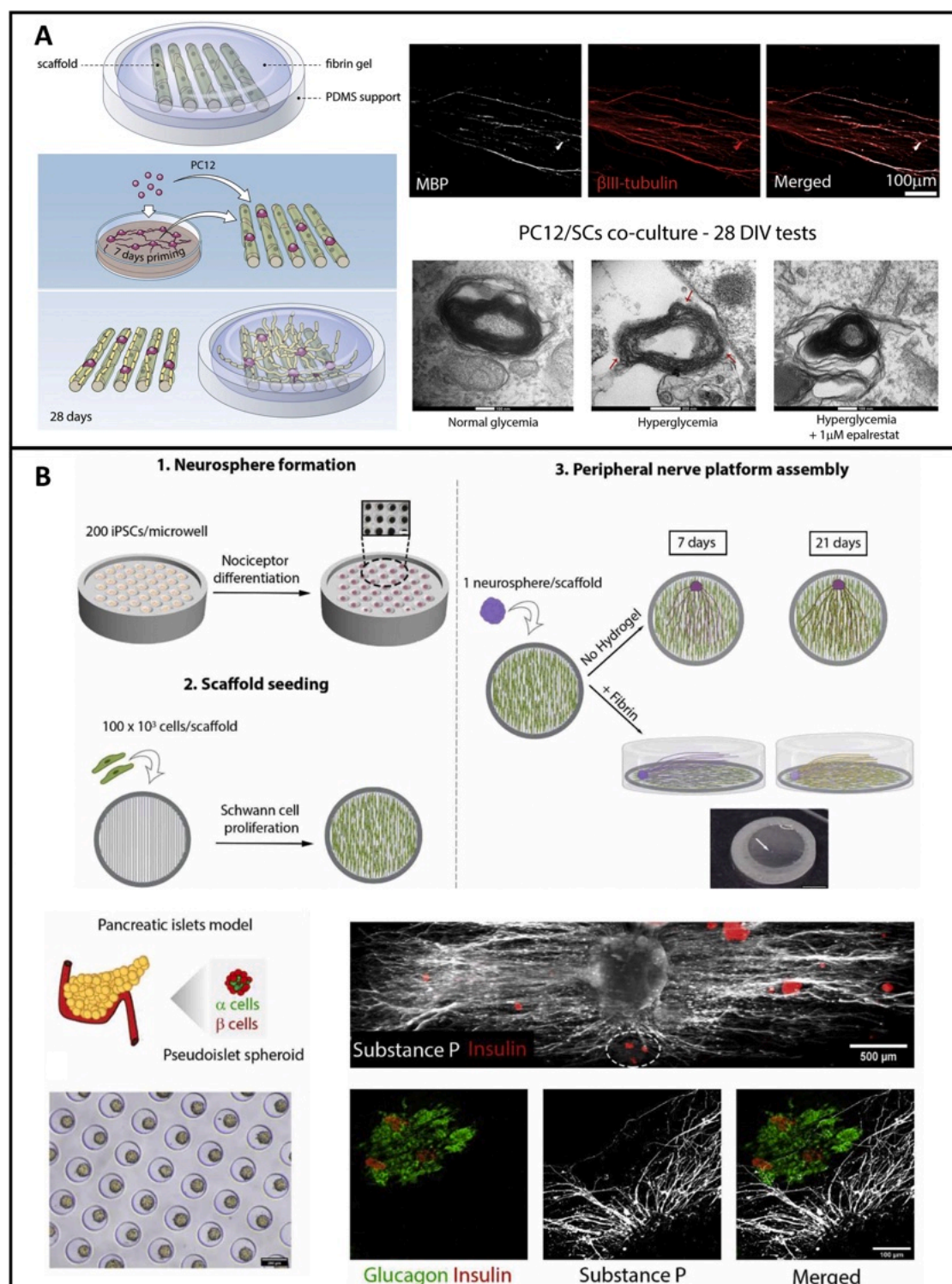


Fig. 7. Scaffold-based physiological and pathological PNS models using 3D printing. (A) A 3D biomimetic PNS model was printed. Bands of Büngner-like structures and Myelination on long-term cultures were successfully formed. Reproduced from [115] under the terms of the Creative Commons CC BY license. (B) The PNS platform was biofabricated in three steps. Pancreatic pseudoislets innervation in the pancreatic islet model, showing infiltrating nociceptor fibers within the pseudoislets. Reproduced from [116] under the terms of the Creative Commons CC BY license.

development for a variety of neural diseases and disorders has been investigated using *in vitro* 3D models. The underlying neurobiological mechanisms responsible for the onset, progression, and treatment of neurodegenerative diseases still remain elusive. The vast majority of (90

% or more) neurodegenerative diseases are not due to genetic origin, making it difficult to generate suitable animal models [117]. The utilization of patient-derived iPSCs in engineered *in vitro* models provides valuable human-specific mechanistic insights in both disease pathology

and therapy.

4.1.1. 3D modeling of neurodegenerative disease

Neurodegenerative diseases such as ADs, PDs, HD, ALS have been widely studied by using 3D models. For example, Ruiz et al. [118] conducted research on microfluidic chips for ADs. They found that the

oligomeric state of A β exhibited greater toxicity compared to the fibrillar form. In neuronal cultures, the exposure to oligomeric A β for 96 h resulted in a fourfold increase in neuronal loss compared to control cultures. In contrast, the toxicity induced by the fibrillar form was significantly lower, with less than a twofold increase observed. Additionally, they studied the effects of the anti-inflammatory drug FTY720

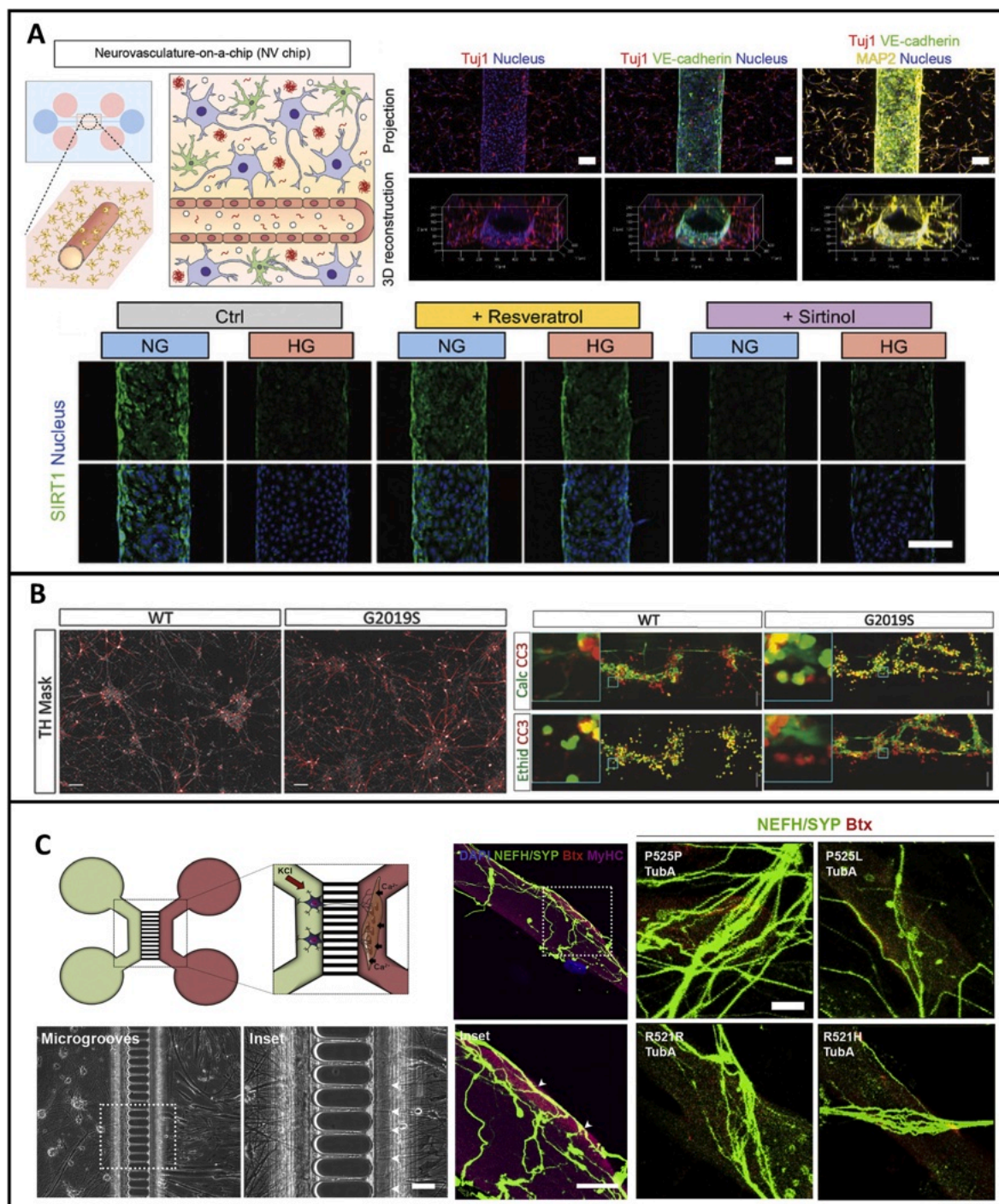


Fig. 8. 3D modeling of neurodegenerative disease. (A) NV chip was fabricated by mimicking the neurovasculature in the diabetic brain. Co-culture of ReN cells and hBMECs and their response were treated with resveratrol or sirtinol under normal and high-glucose conditions. Reproduced from [119] under the terms of the Creative Commons CC BY license. (B) LRRK2-G2019S dependent phenotype was enhanced in 3D conditions, and LRRK2-G2019S carried by PD patients induced mitochondrial abnormalities and cell death in developing neurons. Reproduced from [121] under the terms of the Creative Commons CC BY license. (C) Fabrication of human motor units in microfluidic devices, and the axonal can migrate through microgrooves. ALS-dependent impairments in NMJ morphology can be improved by HDAC6 inhibition [110]. Copyright 2021, Elsevier.

on neuronal cultures and microglia-enriched neuronal cultures, discovering its protective effect. Jang et al. conducted a separate study where they designed a neurovasculature-on-a-chip (NV chip) model that successfully replicated the diabetic microenvironment within the brain. Using the NV chip, the researchers examined the underlying mechanisms of hyperglycemic-induced AD pathogenesis (Fig. 8A) [119]. It was suggested that regulating SIRT1 through pharmaceutical intervention could potentially restore the functionality and molecular state of brain endothelial cells and neurons, thereby halting the progression of hyperglycemia-mediated AD. Zivko et al. used human-induced pluripotent stem cell-derived bulbar organoids to evaluate therapeutic response to NPS of AD. It was found that escitalopram oxalate had different therapeutic effects on organoids in different individuals, which may reflect the heterogeneous responses observed in clinical practice, suggesting that this *in vitro* approach could be used to predict patient response to treatment and to understand the heterogeneity of treatment response [120]. In another research, PD patient-specific neurons that carry the LRRK2-G2019S mutation were cultured on 3D microfluidics [121]. Treatment with the LRRK2 inhibitor 2 (Inh2) was able to rescue LRRK2-G2019S-dependent dopaminergic phenotypes, showing the potential for patient stratification and personalized drug selection (Fig. 8B). Besides, Metzger et al. [122] introduced a novel phenotypic screening methodology using organoids for the detection of modulators of phenotypes of HD. Through this approach, the authors identified certain bromodomain inhibitors that effectively reversed HD phenotypes back to the wild-type (WT) state. Moreover, *in vitro* studies revealed that these inhibitors exhibited the capacity to decrease neuronal susceptibility to apoptosis in human HD neurons.

ALS is distinguished by the deterioration of NMJs followed by muscle atrophy. Numerous studies have focused on developing *in vitro* models of NMJs to identify effective treatment strategies. T. Osaki et al. conducted a study aimed at developing a physiological model of the human motor unit [123]. In the study, non-ALS patient-derived MNs and skeletal muscle cells derived from iPSCs were utilized. The ALS pathological condition was simulated by the addition of excess glutamic acid. Both single treatments and a combination of the inhibitors showed significant effectiveness in preventing the decline in muscle contraction force. These treatments also demonstrated a neuroprotective impact by enhancing autophagy and facilitating the degradation of the TAR DNA binding protein 43 in MNs. Another study by Dittlau et al. developed a versatile and reproducible *in vitro* NMJ model by co-culturing iPSC-derived MNs with myotubes derived from human primary mesoangioblasts in microfluidic devices (Fig. 8C) [110]. The study confirmed the positive effect of the HDAC6 inhibitor Tubastatin A in promoting neurite outgrowth and regeneration.

4.1.2. 3D modeling of neural injuries

The damage or degeneration of nerve tissues leads to human motor dysfunction, which will greatly impair the life quality of patients and also increase social burden. It is urgent to develop nerve regeneration strategies and nerve injury therapies or therapeutic drugs by using 3D models *in vitro*. Xue et al. developed human dorsoventral spinal cord organoids composed of porous chitosan microspheres loaded with spinal cord ventralizing signaling Shh antagonist, presenting a promoting effect for spinal differentiation [124]. Deleglise et al. built up a versatile “brain on a chip” platform to investigate the impact of putative protective molecules on various aspects of local pathological stresses [125]. This platform enabled the examination of somatic, mid-axon, presynaptic, and postsynaptic responses within the brain, providing valuable insights into the effects of these molecules. The findings revealed that NAD⁺ and the Rho-kinase inhibitor Y27632 were effective in preventing synaptic disconnection, while the broad-spectrum caspase inhibitor zVAD-fmk and the stilbenoid resveratrol did not exhibit the same efficacy in preventing presynaptic degeneration. Cultures pretreated with NAD⁺ (5 mM, 24 h) and Rho-kinase inhibitor Y27632 (10 μ M, 1 h pretreatment) retained 65 % and 75 % of the initial number of synapses,

respectively. In contrast, unpretreated cultures only retained 25 % of the initial synapse count. Regarding the regeneration of PNS, a 3D platform that mimicked the PN architecture was established and exposed to diabetic conditions for simulating a pathological state and drug screening. It was shown that epalrestat supplements (1 μ M) could effectively prevent myelin deformation compared to the untreated group [115]. In another study, a reliable microfluidic device was developed to create an *in vitro* model of axon entrapment. This device was utilized to investigate the mechanisms underlying the induction of neurite outgrowth by glial cell-derived neurotrophic factor (GDNF). The results showed that over-expressed levels of GDNF (500–700 ng/mL) alone can significantly cause neurite entrapment [126].

4.1.3. 3D modeling of stroke

Stroke poses a substantial worldwide health burden, holding the second position as a leading cause of mortality and the third position as a cause of disability among adults. Unfortunately, there are limited effective therapies available for stroke patients. At present, the medication approved by the FDA for the treatment of ischemic stroke is Tissue plasminogen activator (tPA), and there is no approved drug for hemorrhagic stroke to the authors' knowledge. The development of effective treatments for stroke is a pressing and critical requirement. In a previous study, researchers established a humanized ischemic stroke model using cerebral organoids (Cos). They tested several neuroprotective agents, including edaravone, butylphthalide, P7C3-A20, and ZL006, at a concentration of 10 μ M each. These agents demonstrated similar beneficial effects for stroke treatment [127]. In another study, three types of astrocytes were identified for various responses to hypoxic environments in a stroke organoid, and a traditional Chinese medicine formula was applied for treating the stroke [128].

A comprehensive understanding of the neurovascular unit (NVU) is essential for gaining deeper insights into ischemic stroke, which occurs due to a temporary or prolonged reduction in cerebral blood flow and the BBB to develop effective treatments. Cecchelli et al. developed a technique to create a human BBB model using hematopoietic stem cells derived from cord blood [129]. They investigated the role of the Wnt/ β -catenin signaling pathway in mediating the BBB inductive properties of pericytes. In a recent study, a hypoxic stroke organoid model was established to simulate ischemia-induced hypoxic stroke. Combining with proteomics, single-cell transcriptomics and histopathological analyses, it was found that three types of astrocytes were significantly increased, showing different responses to the hypoxic environment. Additionally, both *in vivo* and *in vitro* cortical excitatory neurons showed signs of apoptosis and senescence after hypoxia. A traditional Chinese medicine formulation, known as Dengzhan Shengmai Capsule, may have a potential role in the treatment of ischemic and hypoxic stroke by modulating lipid metabolism-related biological functions. Besides, stroke microphysiological systems using organ-on-a-chip technology were also investigated to explore the potential of stem cell therapies for neural recovery. For instance, Lee et al. developed a neurovascular-unit-on-a-chip with human-derived cells, simulating the *in vivo* three-dimensional microenvironment to systematically analyze the neuro-recovery effects of various types of stem cells currently being used in clinical stroke treatments.

4.1.4. 3D modeling of cancer

To effectively treat neural cancers, it is essential to have high-throughput drug screening methods and personalized medicine tailored to individual patients. The development of 3D neural models and chip platforms has provided a rapid, stimuli-sensitive, and customizable approach for efficient drug screening in specific patients. For instance, Fan et al. created a 3D brain cancer chip using photolithography of PDMS and photo-polymerizable poly(ethylene) glycol diacrylate (PEGDA) hydrogel [130]. Glioblastoma cells (U87) were cultured on the chip to form brain cancer tissues, allowing for high-throughput and simultaneous testing. The combination treatment

of pitavastatin and irinotecan was evaluated using this platform.

Organoids have also demonstrated their usefulness for cancer drug testing. A novel model called the 3D heterotypic glioblastoma brain sphere (gBS) model was generated, which consisted of glioblastoma tumor cells, iPSC-derived neurons, glial cells, and astrocytes. This gBS model was employed as an advanced histology platform (microTMA) for efficient drug screening against tumor cells [131]. Treatment with temozolomide (TMZ, 100 μ M) and doxorubicin (DOX, 0.3 μ M) resulted in approximately 30 % ($P \sim 0.07$) and 80 % ($P < 0.001$) decrease in tumor size, respectively, without affecting normal cells. This platform presents an innovative method for screening novel anti-glioblastoma agents and assessing treatment options tailored to individual patients. Ramani et al. induced differentiation directly from hiPSC to neuroepithelium, accurately controlled the size of early neurospheres, and efficiently generated a large number of brain organoids. Patient-derived glioma stem cells (GSCs) were fused with this brain organoid to simulate glioma invasion. Two drugs, Selumetinib and Fulvestrant, were identified through medium-throughput drug screening, which effectively inhibited GSCs invasion of brain organoids *in vitro* and glioma xenografts *in vivo* [132].

4.2. Pharmacokinetics and pharmacodynamics tests on 3D neural models for drug screening

Preclinical screening and testing of the nervous system in 3D models are innovative approaches used in pharmaceutical research to evaluate the pharmacokinetics and pharmacodynamics of potential drug candidates. Pharmacokinetics pertains to the processes by which a drug is absorbed, distributed, metabolized, and eliminated within the body. With 3D nervous system modeling, researchers can simulate drug absorption and distribution based on the unique characteristics of the nervous system, including its cellular structure and tissue composition. Such tests facilitate a deeper comprehension of the interactions between drugs and the nervous system, as well as metabolism in the body. On the other hand, pharmacodynamics focuses on the impacts of a drug on the nervous system. By utilizing 3D modeling, researchers can assess the drug's impact on specific nerve cells, neural networks, and signaling pathways. This allows for the investigation of drug efficacy, target engagement, and potential side effects more realistically and reliably.

Understanding the drug pharmacokinetics and pharmacodynamics in the CNS is particularly important for curing CNS diseases. This is because drug exposure is regulated by the plasma pharmacokinetic properties of the drug, which are influenced by the relationship between the physiological activity of the BBB and the physicochemical properties of the drug molecule. Inefficient metabolism, absorption, excretion, distribution and toxicity of the drug are significant issues for medications to cross the BBB [133]. The BBB plays a regulatory role in maintaining metabolic homeostasis and influencing the pharmacokinetics and pharmacodynamics of drugs [134,135]. Emerging BBB microfluidic models make it possible to control the extracellular microenvironment and fluid delivery, enabling investigations into the toxicity of organophosphates on the BBB. These investigations encompass studying the damage to barrier integrity, inhibition of acetylcholinesterase, and reduction in cellular viability [136]. A range of cell types, such as endothelial cells, neuroblastoma cells, microglia, and astrocytes, have been utilized to study BBB-related toxicity [137]. By functional coupling with chips for other parenchymal organs, such as the liver and kidney, body-on-chip platforms can be generated as comprehensive devices for investigating the sequential metabolism of drugs across multiple organ compartments and thus validate therapeutics for neural diseases [138]. For example, Maoz et al. built three interconnected microfluidic chips to replicate the processes of influx across the BBB, the dynamics within the brain parenchymal compartment, and efflux across the BBB [139]. Leveraging this integrated system, they emulated the impact of the intravascular administration of methamphetamine, a psychoactive substance, and unveiled hitherto unrecognized metabolic interactions

between the BBB and neurons.

In addition, human microphysiological systems offer a promising avenue for modeling the intricate interactions among organs to study the human metabolism resulted from the transport of drugs, metabolites, or xenobiotics [140]. Vernetti et al. [141] scrutinized five human MPS models to replicate the organs pivotal in absorption, metabolism, and clearance processes, namely, the jejunum, liver, and kidney, skeletal muscle and neurovascular models, for their functional coherence and organ-specific processing. The research revealed that trimethylamine-N-oxide has the capability to cross the blood-brain barrier, and demonstrated the potential of human MPS in uncovering multi-organ toxicity and providing insights into absorption, distribution, metabolism, and excretion (ADME) processes. This study shed light on previously undisclosed aspects of pharmacokinetics. In another study, Li et al. contributed a biomimetic liver-brain model designed to replicate both physiological and pharmacological processes *in vitro* [142]. The model offers a straightforward platform conducive to prolonged cell co-culture, drug administration, metabolism assessment, and real-time evaluation of the drug impacts on brain cancer.

Overall, the neural models open a new door for physiological pharmacokinetic and pharmacodynamics modeling, which can accelerate drug development and clinical transformation [143]. However, the reconstruction and integration of blood-tissue and tissue-tissue interfaces into multiorgan chips still present significant technical challenges. It requires a closer collaboration among engineers, biologists, pharmacologists, and computational scientists to overcome these challenges effectively.

4.3. Drug safety evaluation on 3D neural models for preclinical trial

3D neural models bring great expectations to avoid the trial failure such as drug inefficiency or even toxicity due to its ability to replicate the relevant effective drug concentrations observed in humans, surpassing the limitations of 2D models [144]. They also enable the early identification of off-target effects during the drug development process, and minimize the risks and expenditure associated with the development of new drugs. A few previous researches have focused on screening drug efficacy and toxicity using 3D systems. For example, the formation of stable and electrically active neuronal networks on MEA neurochips was achieved by Gramowski et al. through the co-culturing of primary neurons and glial cells, and assessed the neurotoxic effects of nanoparticles including carbon black (CB), hematite (Fe_2O_3), and titanium dioxide (TiO_2) NPs by testing their acute electrophysiological effects [145]. Nierode et al. demonstrated the application of a microarray chip platform for sensitive drug screening [146]. Integrated with 532 microwells for cell culture, the high-throughput system allowed for testing the 3D phenotypic screening and predicting the toxicity of 24 bioactive compounds in human stem/progenitor cells. The study found that there were differences in cell morphology and protein expression between 2D culture and 3D culture. In the case of SOX2, the situation in 2D culture began to decrease after 6 weeks, while in 3D culture it began to increase after 10 days. Similarly, researchers fabricated a cell chip utilizing the electrochemical method to assess the toxic effects of CdSe/ZnS QDs on the intracellular redox environment of neural cells [147]. By comparing the results of toxic effect testing using the conventional MTT assay, it was discovered that the intracellular redox environment exhibited greater sensitivity than other components such as mitochondria and the cell membrane. At the same range of concentrations of Red-TA QDs (5–50 $\mu\text{g/mL}$), the cell viability measured by the MTT assay only decreased by 15 %, whereas the signals obtained from differential pulse voltammetry exhibited a more significant decrease of 37.5 %. The results indicate that the cell chip can be a useful tool for the assessment of cytotoxicity of nanoparticles that have potential toxicity at low concentrations. Wang et al. presented a powerful organ-on-a-chip system with hiPSCs-derived brain organoids to evaluate the effects of prenatal nicotine exposure on neural tissues [148]. Fetal brain

development at early stages can be well reconstructed with clear brain features and the ability of fluid perfusion. The induction of nicotine causes serious neurological disorders such as premature neuronal differentiation, disrupted brain regional organization and abnormal cortical development. Hong et al. created forebrain organoids with a stable and homogeneous expression of channelrhodopsin-2(ChR2) which shows great ability to consistently activate neurons upon light stimulation and induce strong muscle contraction in the hybrid platform where the organoids are connected with spinal cord organoids and skeletal muscle spheroid, giving it a possibility to study neural dynamics and pharmacokinetics in human brains [149].

In summary, nervous system 3D modeling in preclinical trials offers a promising approach for evaluating the safety and efficiency of drug candidates. These models provide insights into the interactions between drugs and the nervous system, enabling researchers to make more informed decisions during the drug development process. However, it is crucial to validate and translate these findings carefully to ensure successful clinical translation.

4.4. Integration of 3D neural models with artificial intelligence

The progression in artificial intelligence (AI) yields notable enhancements for the design and data processing of 3D modeling of neural tissues, especially neural-on-chip approaches. AI-driven visual recognition and data processing hold promise for addressing pertinent

challenges, encompassing optimization of culture conditions, image detection and tracking, and management of vast datasets inherent to OoC research. In a recent investigation, researchers developed a convolutional neural network model to accurately extract essential parameters related to tumor cell behavior from fluorescence images within a glioblastoma OoC framework [150]. This CNN model was subsequently integrated with an *in vitro-in silico* methodology, culminating in the real-time prediction of tumor evolution. In addition, during the preparatory stages of a project, Deep Learning techniques can be utilized for device design and materials selection, enhancing the suitability of OoC systems for tailored applications. While the integration of AI remains limited in OoC research, the existing and emerging body of studies suggests promising avenues for synergistic collaborations between OoCs and AI in the realm of neural drug evaluation.

In conclusion, 3D neural models, including organoids, organ-on-chip systems and 3D-printed scaffolds, enable precise disease modeling for neurodegenerative disorders like AD, PD, and ALS, neural injuries, stroke and cancer, advancing personalized drug testing using patient-derived iPSCs. Pharmacokinetic studies leverage neural models especially BBB-on-chip systems to simulate drug absorption, metabolism, and neurotoxicity, while multi-organ chips assess systemic drug effects. Drug safety assessments in 3D systems can avoid trial failures such as drug inefficiency or even toxicity, outperforming traditional 2D assays. AI-driven platforms analyze tumor evolution in glioblastoma models and optimize drug screening via convolutional neural networks,

Table 1

Summary and comparison of *in vitro* 3D neurological disease models constructed by different methods.

Techniques	Advantages	Limitations	Typical physiological model	Typical disease model	Future development direction
Organoid	- Cell self-assembly and spontaneous differentiation - Structure and function closer to real organs - Reduced reliance on animal testing	- Longer incubation time - Higher experimental costs - Uncontrollable ECM - Usually lacks blood supply - Smaller size compared to human organs - Difficulty in reproducing highly complex organs	- CNS - Cortical organoids [29, 30] - Spinal cord organoid [7, 34,36,155] - PNS - Cochlear organoids [87] - Sensory ganglion organoids [88] - Dorsal root ganglion (DRG) organoid [89] - Neuro-mesodermal assembloid [90]	- CNS - Spinal cord injury [7,155] - Depressive disorder [38] - Amyotrophic lateral sclerosis [41,156] - Alzheimer's disease [35,44] - Parkinson's disease [45] - Brain cancers [48] - PNS - Artificial cochlea [87]	- Multi-organ system modeling for studying organ interactions - Combining with biochip and microfluidics for improving real-time organoid monitoring capabilities - Integrating organoid experimental data with computational modeling for a deeper understanding of organ structure, function, and response - Personalized disease research and drug development
Organ-on-a-chip	- Precise control of cell growth environment - Parallel study of multiple organs or tissues <i>In situ</i> stimulation and monitoring - High-throughput and efficient drug screening - Reduced reliance on animal testing	- Low structural complexity compared to organoid - Long-term stability and reliability to be improved - Lack of standardized methods and guidelines	- CNS - Synapse-on-a-chip [52] - Cortico-striatal connectivity [53] - Myelin-on-chip [54] - BBB-on-Chips [60,56] - PNS - Neurite outgrowth models [93] - Motor neuron model [94] - Nerve-on-chip platform [96]	- CNS - Spinal cord injury [55] - Blood brain barrier [60,61] - Alzheimer's disease [57] - Parkinson's disease [65] - Huntington's disease [94] - PNS - Peripheral nerve injury [96] - Demyelinating diseases [102]	- Multi-organ chip association to simulate the functions and interactions of human systems - Combined with machine learning and artificial intelligence to deepen understanding of drug responses and disease mechanisms - Personalized disease modeling and drug screening
3D printing	- Layer-by-layer modulation of cell position at the microscopic scale - Personalized neural model construction	- Long-term stability and viability of cells to be improved - Difficulty in accurately reproducing subtle neural structures and functions	- CNS - Lipid-bilayer-supported 3D printing for human cerebral cortex development [82] - PNS - Composite scaffold loaded with growth factor for neural regeneration [114] 3D oriented scaffolds for neural Myelination [115]	- CNS - Spinal cord injury [80,81] - Blood brain barrier [84]	- Higher resolution printing capabilities - Construct multi-scale neural models to comprehensively study the hierarchical structure and functions of neural networks - Combining biosensing and microfluidic technology to achieve more realistic simulation of neural activity

enhancing predictive accuracy. Challenges remain in replicating full neural complexity and integrating multi-organ interactions, but advancements in bioprinting, vascularization, and AI integration promise accelerated physiologically relevant drug discovery. These innovations bridge preclinical and clinical gaps, reducing trial failures and enabling tailored therapies for neurological and oncological disorders.

5. Challenges and conclusions remark

The benefits, constraints, and prospective research avenues of *in vitro* 3D neural models constructed by different methods are compared in Table 1. As single use of each approach has its unavoidable limitations, integrating organoids, OoCs, and 3D printed scaffolds into hybrid models is a promising strategy to address the individual shortcomings of these systems. For instance, cerebral organoids can be engineered with brain-like ECM to recreate brain-mimetic niches to guide neural and glial differentiation, which would be difficult in the non-neuronal matrix [151]. Furthermore, microfluidic flow can be integrated with the organoids to facilitate the oxygen supply and nutrient/waste exchange, thus reducing cell death during organoid development [151,152]. In addition, the convergence of 3D bioprinting and OoCs can create more complex artificial tissues with the proper microarchitecture for mechanical and chemical stimuli [153]. 3D printing can also be applied to spatially patterned organoids with vascular/glial cells to enhance cell diversity in neural models and improve reproducibility by controlling cell placement, which could not be obtained by single organoids approach [154]. Another future direction is to build hybrid models including neural organoids with other organ chips (e.g., liver, immune system) on a multi-organ-on-chip to investigate systemic organ interactions and predict systemic drug effects.

Current 3D models of the nervous system, while promising platforms for drug evaluation, face persistent challenges and critical limitations that hinder their clinical translation. First, it is still challenging to replicate the complexity of the nervous system accurately in experimental 3D systems. Materials and manufacturing techniques need to be improved to construct nerve-mimetic models representing various components, interactions, and functions of different cell types, neurons, and synapses of the intricate neural structure. Second, despite significant advancements in neuroscience, our understanding of the nervous system is still limited. There are several unknowns, such as the exact mechanisms underlying certain neurological disorders or the specific ways in which drugs interact with the nervous system. This lack of understanding makes it difficult to create comprehensive and accurate 3D models. Third, it is essential to ensure that these models can accurately represent the real-life behavior of the nervous system and predict the effects of drugs accurately. Developing reliable and validated 3D models for the nervous system is crucial for drug evaluation. Achieving this level of reliability requires extensive experimental data and validation studies, which can be time-consuming and resource-intensive. Fourth, most of the evaluation indicators of existing *in vitro* nerve models are based on cell or molecular level detection, and there is not a close enough correlation with the nerve function recovery indicators observed in clinical practice (such as limb motor function, sensory function, etc.), and they are unable to fully represent the therapeutic effects of drugs in clinical settings. In addition, drugs with neuroprotective or reparative effects screened *in vitro* may encounter various problems such as pharmacokinetics, drug toxicity, and immune responses *in vivo*. For example, drugs that can effectively promote the growth and repair of nerve cells *in vitro* may fail to achieve the expected therapeutic effects *in vivo* due to reasons such as difficulty in reaching the injured site, rapid metabolism, or the induction of adverse reactions. Currently, the screening and evaluation of *in vitro* models need to be further clarified and refined to ensure the safety and effectiveness of drug clinical translation.

3D neural models can not only aid in the discovery and development of new drugs but also in the optimization of existing ones. Through the simulation of drug effects on nervous system functions, researchers can

effectively identify potential therapeutic targets and assess the effectiveness of various drug formulations [157]. Time and costs associated with traditional trial-and-error-based drug development approaches can be reduced by applying 3D models. Among all those 3D models, OoC systems stand out due to their potential to enable earlier stages of drug discovery and development through parallelization and automation for high-throughput (HT) drug tests [158]. Although a few HT OoC systems have already been advanced by either increasing the number of replicates per chip [159,160], or injection of cell-laden hydrogels instead of cell suspensions for parallelized [161], remaining challenges still exist to progress HT OoC systems. For example, miniaturized multi-organ platforms in microplate formats or interchangeable organ modules should be designed and fabricated with standardized protocols to ensure reproducibility and compatibility with existing screening infrastructure for scalable production. Furthermore, automation of cell seeding, drug dosing, and media changes, combined with embedded sensors for real-time monitoring of metabolic or physiological parameters (e.g., oxygen, pH, neurophysiology), can enhance throughput and reduce reliance on labor-intensive endpoint assays. Advanced tissue engineering strategies are also unelectable such as improving the physiological relevance while maintaining the scalability of iPSC-derived patient-specific cells, 3D bioprinted organoids, and vascularized networks. Therefore, further comprehensive collaborative efforts including engineering and biological challenges should be solved to bridge the gap between physiologically relevant OoC models and high-throughput scalability.

Regarding the direction of drug development and clinical transformation, there are a few important trends: first, the focus of drug development is shifting towards personalized medicine, where treatments are customized to match an individual's distinct genetic profile and specific neurological condition. For example, using the patient's own induced pluripotent stem cells to build a dedicated 3D model of the nervous system, simulate their specific nervous system diseases, such as Alzheimer's disease, Parkinson's disease, etc., deeply analyze the pathogenesis and progression mechanism, and select the most suitable drugs based on individual differences to achieve personalized drug screening and efficacy prediction. 3D models can contribute to this approach by providing a platform to simulate and evaluate drug responses specific to an individual's nervous system. At the same time, combining the patient's medical imaging data with the 3D model of the nervous system can also provide accurate preoperative planning for neurosurgery, determine the optimal surgical path, and improve the success rate and safety of surgery. In addition, deep integration of the 3D model of the nervous system with the 3D models of other organs can construct multi-organ assemblies to simulate the complex physiological systems of the human body and the interactions between organs. For example, the neural-cardiovascular-liver assembly can explore the regulation of the nervous system on cardiovascular and liver functions and the pathophysiological associations between organs, and help study the mechanisms of systemic diseases and develop treatment strategies. In drug development and toxicology research, it can be used to evaluate the effects of drugs on multiple organs and the metabolic processes in the body, and predict efficacy and toxic side effects. Second, the notable progress made in molecular biology and neuroscience has facilitated the identification of specific targets within the nervous system for drug interventions. 3D models help researchers design and develop drugs that specifically target these identified molecular pathways, potentially leading to more effective and precise treatments for neurological disorders. Third, drug development is increasingly becoming a data-driven field, and the integration of various types of data, such as genomics, proteomics, neuroimaging, and clinical data, is crucial. 3D models can integrate and analyze these different data sources, providing a more comprehensive understanding of drug mechanisms and patient responses. Fourth, the use of artificial intelligence (AI) and machine learning algorithms in drug development is growing rapidly. These technologies can analyze large datasets, identify patterns, accelerate

drug discovery, and optimize treatment strategies [162,163]. The combination of the 3D model of the nervous system and AI can quickly screen a large number of drug molecules and predict their activity, toxicity and pharmacokinetic characteristics in the nervous system. AI helps optimize drug structure, improve the success rate of research and development, and reduce costs by analyzing the drug action data presented by the 3D model. In addition, with the help of patient-specific 3D models of the nervous system and AI, personalized drug screening and efficacy prediction can be achieved, providing strong support for precision medicine and promoting the development of neurological drugs to a new level. Overall, nervous system 3D models for drug evaluation face challenges in accurately representing the complexity of the nervous system. However, with advancements in personalized medicine, targeted therapies, data integration, and AI, there is an exciting direction for drug development and clinical transformation in the field of neuroscience.

Looking toward the future, nervous system 3D models will continue to play a crucial role in drug development [164]. As computational power and modeling techniques improve, advancements in AI and machine learning can be applied to analyze and interpret the vast amounts of data generated by experimental 3D models. Integration of multiple levels of modeling, including molecular, cellular, and tissue scales, will contribute to a comprehensive understanding of the nervous system and its response to drugs.

In conclusion, nervous system 3D modeling holds great promise for the future of drug evaluation. These models serve as a vital tool for evaluating drug safety and efficiency, enabling researchers to make well-informed decisions throughout the drug development process. In the future, by utilizing 3D neural models to accurately represent individual patient characteristics and recapitulate disease pathology *in vitro*, researchers can screen potential drug candidates more efficiently and identify novel therapeutic targets. The integration of emerging technologies such as microfluidics, robotics, advanced imaging techniques and CRISPR/Cas9 genome editing into 3D neural models would help researchers facilitate the swift screening of extensive compound libraries to identify promising drug candidates for further investigation. Further, machine learning algorithms and computational modeling techniques can be combined with 3D neural models to predict drug responses and optimize treatment regimens. Last but not least, the predictive validity of these models must be established and is essential for translating experimental results into clinically relevant therapies for neurological disorders. As technological advancements persist and the field matures, nervous system 3D modeling will assume an increasingly crucial role in shaping the future of medicine, harboring immense potential to revolutionize the field and enhance patient outcomes, ultimately leading to safer and more effective treatments for neurological disorders.

CRedit authorship contribution statement

Hui Zhu: Writing – review & editing, Writing – original draft, Conceptualization. **Cong Yao:** Writing – review & editing. **Zhengqi Xu:** Resources, Methodology. **Guojin Shang:** Resources, Methodology. **Jianhua Peng:** Visualization. **Huangfan Xie:** Visualization. **Tingyu Qian:** Visualization. **Zhennan Qiu:** Writing – review & editing. **Lidia Maeso:** Writing – review & editing. **Mao Mao:** Writing – review & editing. **Yucheng Liao:** Writing – original draft, Conceptualization. **Yong Jiang:** Supervision, Project administration, Funding acquisition. **Dichen Li:** Supervision, Project administration. **Gorka Orive:** Supervision, Project administration, Writing – review & editing. **Aldo R. Bocaccini:** Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] R. Kuner, Central mechanisms of pathological pain, *Nat. Med.* 16 (2010) 1258–1266.
- [2] J. Pakpoor, M. Goldacre, The increasing burden of mortality from neurological diseases, *Nat. Rev. Neurol.* 13 (9) (2017) 518–519.
- [3] M. De Rycker, S. Wyllie, D. Horn, K.D. Read, I.H. Gilbert, Anti-trypanosomatid drug discovery: progress and challenges, *Nat. Rev. Microbiol.* 21 (1) (2023) 35–50.
- [4] P. Khoshkhalagh, A. Sivakumar, L.A. Pace, D.W. Sazer, M.J. Moore, Methods for fabrication and evaluation of a 3D microengineered model of myelinated peripheral nerve, *J. Neural Eng.* 15 (6) (2018) 11.
- [5] D.E. Pankevich, B.M. Altevogt, J. Dunlop, F.H. Gage, S.E. Hyman, Improving and accelerating drug development for nervous system disorders, *Neuron* 84 (3) (2014) 546–553.
- [6] K.-H. Nam, A.S. Smith, S. Lone, S. Kwon, D.-H. Kim, Biomimetic 3D tissue models for advanced high-throughput drug screening, *J. Lab. Autom.* 20 (3) (2015) 201–215.
- [7] J. Xu, S. Fang, S. Deng, H. Li, X. Lin, Y. Huang, S. Chung, Y. Shu, Z. Shao, Generation of neural organoids for spinal-cord regeneration via the direct reprogramming of human astrocytes, *Nat. Biomed. Eng.* 7 (3) (2023) 253–269.
- [8] Y.K. Adlakha, Human 3D brain organoids: steering the demolecularization of brain and neurological diseases, *Cell Death. Discov.* 9 (1) (2023) 221.
- [9] X. Wang, Y. Luo, Y. Ma, P. Wang, R. Yao, Converging bioprinting and organoids to better recapitulate the tumor microenvironment, *Trends Biotechnol.* 42 (5) (2024) 648–663.
- [10] Y.J. Kang, Y.N. Diep, M. Tran, V.T.A. Tran, G. Ambrin, H. Ngo, H. Cho, Three-dimensional human neural culture on a chip recapitulating neuroinflammation and neurodegeneration, *Nat. Protoc.* 18 (9) (2023) 2838–2867.
- [11] A. Sahasrabudhe, L.E. Rupprecht, S. Orguc, T. Khudiyev, T. Tanaka, J. Sands, W. Zhu, A. Tabet, M. Manthey, H. Allen, Multifunctional microelectronic fibers enable wireless modulation of gut and brain neural circuits, *Nat. Biotechnol.* (2023) 1–13.
- [12] Y. Yan, X. Li, Y. Gao, S. Mathivanan, L. Kong, Y. Tao, Y. Dong, X. Li, A. Bhattacharyya, X. Zhao, 3D bioprinting of human neural tissues with functional connectivity, *Cell Stem Cell* 31 (2) (2024) 260–274, e7.
- [13] S.M. Willerth, Bioprinting functional neural networks, *Cell Stem Cell* 31 (2) (2024) 151–152.
- [14] W. Xue, W. Shi, Y. Kong, M. Kuss, B. Duan, Anisotropic scaffolds for peripheral nerve and spinal cord regeneration, *Bioact. Mater.* 6 (11) (2021) 4141–4160.
- [15] Z. Zhao, X. Chen, A.M. Dowbaj, A. Sljukic, K. Bratlie, L. Lin, E.L.S. Fong, G. M. Balachander, Z. Chen, A. Soragni, Organoids, *Nat. Rev. Methods Primers.* 2 (1) (2022) 94.
- [16] C.M. Leung, P. De Haan, K. Ronaldson-Bouchard, G.-A. Kim, J. Ko, H.S. Rho, Z. Chen, P. Habibovic, N.L. Jeon, S. Takayama, A guide to the organ-on-a-chip, *Nat. Rev. Methods Primers.* 2 (1) (2022) 33.
- [17] Y.N. Kang, N. Chou, J.-W. Jang, H.K. Choe, S. Kim, A 3D flexible neural interface based on a microfluidic interconnection cable capable of chemical delivery, *Microsyst. Nanoeng.* 7 (1) (2021) 66.
- [18] H. Zhu, C. Yao, B. Wei, C. Xu, X. Huang, Y. Liu, J. He, J. Zhang, D. Li, 3D printing of functional bioengineered constructs for neural regeneration: a review, *Int. J. Extreme Manuf.* (2023).

- [19] A. Żuchowska, P. Baranowska, M. Flont, Z. Brzózka, E. Jastrzębska, Review: 3D cell models for organ-on-a-chip applications, *Anal. Chim. Acta* 1301 (2024) 342413.
- [20] V.K. Gribkoff, L.K. Kaczmarek, The need for new approaches in CNS drug discovery: why drugs have failed, and what can be done to improve outcomes, *Neuropharmacology* 120 (2017) 11–19.
- [21] R. Rauti, N. Renous, B.M. Maoz, Mimicking the brain extracellular matrix *in vitro*: a review of current methodologies and challenges, *Isr. J. Chem.* 60 (12) (2020) 1141–1151.
- [22] B. Maoz, Brain-on-a-chip: characterizing the next generation of advanced *in vitro* platforms for modeling the central nervous system, *APL Bioeng.* 5 (3) (2021) 14.
- [23] A. Padhi, A.S. Nain, ECM in differentiation: a review of matrix structure, composition and mechanical properties, *Ann. Biomed. Eng.* 48 (3) (2020) 1071–1089.
- [24] O. Chaudhuri, L. Gu, D. Klumpers, M. Darnell, S.A. Bencherif, J.C. Weaver, N. Huebsch, H.P. Lee, E. Lippens, G.N. Duda, D.J. Mooney, Hydrogels with tunable stress relaxation regulate stem cell fate and activity, *Nat. Mater.* 15 (3) (2016) 326. –+.
- [25] D. Dutta, I. Heo, H. Clevers, Disease modeling in stem cell-derived 3D organoid systems, *Trends. Mol. Med.* 23 (5) (2017) 393–410.
- [26] A.D. Rubio, L. Hamilton, M. Bausch, M. Jin, A. Papetti, P. Jiang, S. V. Yelamanchili, A comprehensive review on utilizing Human brain organoids to study neuroinflammation in neurological disorders, *J. Neuroimmune Pharmacol.* 20 (1) (2025) 23.
- [27] A. Uzquiano, P. Arlotta, Brain organoids: the quest to decipher human-specific features of brain development, *Curr. Opin. Genet. Dev.* 75 (2022) 8.
- [28] G. Zhou, S. Pang, Y. Li, J. Gao, Progress in the generation of spinal cord organoids over the past decade and future perspectives, *Neural Regen. Res.* 19 (5) (2024) 1013–1019.
- [29] A. Gordon, S.J. Yoon, S.S. Tran, C.D. Makinson, J.Y. Park, J. Andersen, A. M. Valencia, S. Horvath, X.S. Xiao, J.R. Huguenard, S.P. Pasca, D.H. Geschwind, Long-term maturation of human cortical organoids matches key early postnatal transitions, *Nat. Neurosci.* 24 (3) (2021) 331–342.
- [30] M.A. Cadena, A. Sing, K. Taylor, L.Q. Jin, L.Q. Ning, M.S. Amoli, Y. Singh, S. N. Lanjewar, M.L. Tomov, V. Serpooshan, S.A. Sloan, A 3D bioprinted cortical organoid platform for modeling Human brain development, *Adv. Healthc. Mater.* 13 (27) (2024).
- [31] F. Birey, M.Y. Li, A. Gordon, M.V. Thete, A.M. Valencia, O. Revah, A.M. Pasca, D. H. Geschwind, S.P. Pasca, Dissecting the molecular basis of human interneuron migration in forebrain assemblies from Timothy syndrome, *Cell Stem Cell* 29 (2) (2022) 248. –+.
- [32] S.L. Giandomenico, S.B. Mierau, G.M. Gibbons, L.M.D. Wenger, L. Masullo, T. Sit, M. Sutcliffe, J. Boulanger, M. Tripodi, E. Derivery, O. Paulsen, A. Lakatos, M. A. Lancaster, Cerebral organoids at the air-liquid interface generate diverse nerve tracts with functional output, *Nat. Neurosci.* 22 (4) (2019) 669. –+.
- [33] B. Paulsen, S. Velasco, A.J. Kedaigle, M. Pignoni, G. Quadrato, A.J. Deo, X. Adiconis, A. Uzquiano, R. Sartore, S.M. Yang, S.K. Simmons, P. Symvoulidis, K. Kim, K. Tsafou, A. Podury, C. Abbate, A. Tucewicz, S.N. Smith, A. Albanese, L. Barrett, N.E. Sanjana, X. Shi, K. Chung, K. Lage, E.S. Boyden, A. Regev, J. Z. Levin, P. Arlotta, Autism genes converge on asynchronous development of shared neuron classes, *Nature* 602 (7896) (2022) 268. –+.
- [34] Z.J. Khong Winanto, B.S. Soh, Y. Fan, S.Y. Ng, Organoid cultures of MELAS neural cells reveal hyperactive notch signaling that impacts neurodevelopment, *Cell Death. Dis.* 11 (3) (2020).
- [35] J. Zhao, Y. Fu, Y. Yamazaki, Y. Ren, M.D. Davis, C.C. Liu, W. Lu, X. Wang, K. Chen, Y. Cherukuri, L. Jia, Y.A. Martens, L. Job, F. Shue, T.T. Nguyen, S. G. Younkun, N.R. Graff-Radford, Z.K. Wszolek, D.A. Brafman, Y.W. Asmann, N. Ertekin-Taner, T. Kanekiyo, G. Bu, APOE4 exacerbates synapse loss and neurodegeneration in Alzheimer's disease patient iPSC-derived cerebral organoids, *Nat. Commun.* 11 (1) (2020) 5540.
- [36] W. Xue, B. Li, H. Liu, Y. Xiao, B. Li, L. Ren, H. Li, Z. Shao, Generation of dorsoventral human spinal cord organoids via functionalizing composite scaffold for drug testing, *iScience* 26 (1) (2023) 105898.
- [37] F.K. Liu, J. Huang, L.Y. Zhang, J.D. Chen, Y. Zeng, Y.J. Tang, Z.X. Liu, Advances in cerebral organoid systems and their application in disease modeling, *Neuroscience* 399 (2019) 28–38.
- [38] Y. Li, P.-M. Zeng, J. Wu, Z.-G. Luo, Advances and applications of brain organoids, *Neurosci. Bull.* 39 (11) (2023) 1703–1716.
- [39] A. Kathuria, K. Lopez-Lengowski, S.S. Jagtap, D. McPhie, R.H. Perlis, B.M. Cohen, R. Karmacharya, Transcriptomic landscape and functional characterization of induced pluripotent stem cell-derived cerebral organoids in schizophrenia, *JAMA Psychiatry* 77 (7) (2020) 745–754.
- [40] C. Zivko, R. Sagar, A. Xydia, A. Lopez-Montes, J. Mintzer, P.B. Rosenberg, D. M. Shade, A.P. Porsteinsson, C.G. Lyketsos, V. Mahairaki, iPSC-derived hindbrain organoids to evaluate escitalopram oxalate treatment responses targeting neuropsychiatric symptoms in Alzheimer's disease, *Mol. Psychiatry* 29 (11) (2024) 3644–3652.
- [41] J.D. Pereira, D.M. DuBreuil, A.C. Devlin, A. Held, Y. Sapir, E. Berezovsky, J. Hawrot, K. Dorfman, V. Chandler, B.J. Wainger, Human sensorimotor organoids derived from healthy and amyotrophic lateral sclerosis stem cells form neuromuscular junctions, *Nat. Commun.* 12 (1) (2021) 4744.
- [42] J.-M. Faustino Martins, C. Fischer, A. Urzi, R. Vidal, S. Kunz, P.-L. Ruffault, L. Kabuss, I. Hube, E. Gazziero, C. Birchmeier, S. Spuler, S. Sauer, M. Gouti, Self-organizing 3D Human trunk neuromuscular organoids, *Cell Stem Cell* 26 (2) (2020) 172–186, e6.
- [43] J. Zhao, Y. Fu, Y. Yamazaki, Y.X. Ren, M.D. Davis, C.C. Liu, W.Y. Lu, X. Wang, K. Chen, Y. Cherukuri, L. Jia, Y.A. Martens, L. Job, F. Shue, T.T. Nguyen, S. G. Younkun, N.R. Graff-Radford, Z.K. Wszolek, D.A. Brafman, Y.W. Asmann, N. Ertekin-Taner, T. Kanekiyo, G.J. Bu, APOE4 exacerbates synapse loss and neurodegeneration in Alzheimer's disease patient iPSC-derived cerebral organoids, *Nat. Commun.* 11 (2021) 5540, 12(1) (2021) 1.
- [44] X. Chen, G. Sun, E. Tian, M. Zhang, H. Davtyan, T.G. Beach, E.M. Reiman, M. Blurton-Jones, D.M. Holtzman, Y. Shi, Modeling sporadic Alzheimer's disease in Human brain organoids under serum exposure, *Adv. Sci.* 8 (18) (2021) e2101462.
- [45] N. Wulansari, W.H.W. Darsono, H.J. Woo, M.Y. Chang, J. Kim, E.J. Bae, W. Sun, J.H. Lee, L.J. Cho, H. Shin, S.J. Lee, S.H. Lee, Neurodevelopmental defects and neurodegenerative phenotypes in human brain organoids carrying Parkinson's disease-linked DNAJC6 mutations, *Sci. Adv.* 7 (8) (2021).
- [46] A. Linkous, D. Balamatsias, M. Snuderl, L. Edwards, K. Miyaguchi, T. Milner, B. Reich, L. Cohen-Gould, A. Storaska, Y. Nakayama, E. Schenkein, R. Singhanian, S. Cirigliano, T. Magdeldin, Y. Lin, G. Nanjangud, K. Chadalavada, D. Pisapia, C. Liston, H.A. Fine, Modeling patient-derived glioblastoma with cerebral organoids, *Cell Rep.* 26 (12) (2019) 3203. –+.
- [47] Y. Wu, J. Cheng, J. Qi, C. Hang, R. Dong, B.C. Low, H. Yu, X. Jiang, Three-dimensional liquid metal-based neuro-interfaces for human hippocampal organoids, *Nat. Commun.* 15 (1) (2024).
- [48] S. Bian, M. Repic, Z. Guo, A. Kavirayani, T. Burkard, J.A. Bagley, C. Krauditsch, J. A. Knoblich, Genetically engineered cerebral organoids model brain tumor formation, *Nat. Methods* 15 (8) (2018) 631–639.
- [49] D. Hendriks, A. Pagliaro, F. Andreatta, Z. Ma, J. van Giessen, S. Massalini, C. López-Iglesias, G.J.F. van Son, J. DeMartino, J.M.A. Damen, I. Zoutendijk, N. Staliarova, A.L. Bredenoord, F.C.P. Holstege, P.J. Peters, T. Margaritis, S. Chuva de Sousa Lopes, W. Wu, H. Clevers, B. Artegiani, Human fetal brain self-organizes into long-term expanding organoids, *Cell* 187 (3) (2024) 712–732, e38.
- [50] T. Peng, X. Ma, W. Hua, C. Wang, Y. Chu, M. Sun, Y. Fermi, S. Hamelmann, K. Lindner, C. Shao, J. Zaman, W. Tian, Y. Zhuo, Y. Harim, N. Stöfler, L. Hammann, Q. Xiao, X. Jin, R. Warta, C. Lotsch, X. Zhuang, Y. Feng, M. Fu, X. Zhang, J. Zhang, H. Xu, F. Qiu, L. Xie, Y. Zhang, W. Zhu, Z. Du, L. Salgueiro, M. Schneider, F. Eichhorn, A. Lefevre, S. Pusch, V. Grinevich, M. Ratliff, S. Loges, L. Bunse, F. Sahm, Y. Xiang, A. Unterberg, A. von Deimling, M. Platten, C. Herold-Mende, Y. Wu, H.-K. Liu, Y. Mao, Individualized patient tumor organoids faithfully preserve human brain tumor ecosystems and predict patient response to therapy, *Cell Stem Cell*.
- [51] J.F. Li, Y.T. Dai, H. Lilljebjörn, S.H. Shen, B.W. Cui, L. Bai, Y.F. Liu, M.X. Qian, Y. Kubota, H. Kiyoi, I. Matsumura, Y. Miyazaki, L. Olsson, A.M. Tan, H. Ariffin, J. Chen, J. Takita, T. Yasuda, H. Mano, B. Johansson, J.J. Yang, A.E.J. Yeoh, F. Hayakawa, Z. Chen, C.H. Pui, T. Fioretto, S.J. Chen, J.Y. Huang, Transcriptional landscape of B cell precursor acute lymphoblastic leukemia based on an international study of 1223 cases, in: *Proceedings of the National Academy of Sciences of the United States of America* 115, 2018, pp. E11711–E11720.
- [52] L. Amirifar, A. Shamloo, R. Nasiri, N.R. de Barros, Z.Z. Wang, B.D. Unluturk, A. Libanori, O. Ievlevskiy, S.E. Dilemmez, S. Sances, I. Balasingham, S.K. Seidlits, N. Ashammakhi, Brain-on-a-chip: recent advances in design and techniques for microfluidic models of the brain in health and disease, *Biomaterials* 285 (2022) 121531.
- [53] F. Nebuloni, Q.B. Do, P.R. Cook, E.J. Walsh, R. Wade-Martins, A fluid-walled microfluidic platform for human neuron microcircuits and directed axotomy, *Lab. Chip.* 24 (13) (2024) 3252–3264.
- [54] B.E. Kerman, H.J. Kim, K. Padmanabhan, A. Mei, S. Georges, M.S. Joens, J. A. Fitzpatrick, R. Jappelli, K.J. Chandross, P. August, F.H. Gage, *In vitro* myelin formation using embryonic stem cells, *Development* 142 (12) (2015) 2213–2225.
- [55] P. Mohammadi, S. Nadri, A. Abdanipour, Y. Mortazavi, Microchip encapsulation and microRNA-7 overexpression of trabecular meshwork mesenchymal stem/stromal cells improve motor function after spinal cord injury, *J. Biomed. Mater. Res. A* 111 (9) (2023) 1482–1494.
- [56] M. Campisi, Y. Shin, T. Osaki, C. Hajal, V. Chiono, R.D. Kamm, 3D self-organized microvascular model of the human blood-brain barrier with endothelial cells, pericytes and astrocytes, *Biomaterials* 180 (2018) 117–129.
- [57] J. Park, B.K. Lee, G.S. Jeong, J.K. Hyun, C.J. Lee, S.H. Lee, Three-dimensional brain-on-a-chip with an interstitial level of flow and its application as an *in vitro* model of Alzheimer's disease, *Lab. Chip.* 15 (1) (2015) 141–150.
- [58] B. Peng, S.P. Hao, Z.Q. Tong, H. Bai, S.J. Pan, K.L. Lim, L. Li, N.H. Voelcker, W. Huang, Blood-brain barrier (BBB)-on-a-chip: a promising breakthrough in brain disease research, *Lab. Chip.* 22 (19) (2022) 3579–3602.
- [59] A. Refaat, P. Thomas, W. Zhang, L. Esser, Z. Tong, M. Beer, D. Mawdsley, D. Thirkettle-Watts, K.A. Shields, J.A. Nicolazzo, N.H. Voelcker, An *In vitro-In vivo* comparative study using highly sensitive radioisotopic assays to assess the predictive power of emerging blood-brain barrier models, *Small. Methods* (2024).
- [60] S.I. Ahn, Y.J. Sei, H.J. Park, J. Kim, Y. Ryu, J.J. Choi, H.J. Sung, T.J. MacDonald, A.I. Levey, Y. Kim, Microengineered human blood-brain barrier platform for understanding nanoparticle transport mechanisms, *Nat. Commun.* 11 (1) (2020) 175.
- [61] Z. Lyu, J. Park, K.M. Kim, H.J. Jin, H. Wu, J. Rajadas, D.H. Kim, G.K. Steinberg, W. Lee, A neurovascular-unit-on-a-chip for the evaluation of the restorative potential of stem cell therapies for ischaemic stroke, *Nat. Biomed. Eng.* 5 (8) (2021) 847–863.
- [62] C.S. Ahuja, S. Nori, L. Tetreault, J. Wilson, B. Kwon, J. Harrop, D. Choi, M. G. Fehlings, Traumatic spinal cord injury-repair and regeneration, *Neurosurgery* 80 (3s) (2017) S9–S22.

- [63] T. Nagendran, A.M. Taylor, Unique axon-to-soma signaling pathways mediate dendritic spine loss and hyper-excitability post-axotomy, *Front. Cell Neurosci.* 13 (2019) 8.
- [64] S. Sun, J. Kanagaraj, L. Cho, D. Kang, S. Xiao, M. Cho, Characterization of subcellular responses induced by exposure of microbubbles to astrocytes, *J. Neurotrauma* 32 (19) (2015) 1441–1448.
- [65] J.A. Kim, S. Hong, W.J. Rhee, Microfluidic three-dimensional cell culture of stem cells for high-throughput analysis, *World J. Stem Cells* 11 (10) (2019) 803–816.
- [66] A. Virlogeux, E. Moutaux, W. Christaller, A. Genoux, J. Bruyère, E. Fino, B. Charlot, M. Cazorla, F. Saudou, Reconstituting corticostriatal network on-a-chip reveals the contribution of the presynaptic compartment to Huntington's disease, *Cell Rep.* 22 (1) (2018) 110–122.
- [67] S. Li, C. Huang, H. Liu, Z. Wang, X. Han, Z. Chen, J. Huang, Z. Wang, Optimization of cellular microenvironment of custom 3D printed brain-mimetic hydrogel and its application in drug neurotoxicity evaluation, *Biomater. Adv.* 172 (2025) 214257.
- [68] H. Zhu, H. Kuang, X. Huang, X. Li, R. Zhao, G. Shang, Z. Wang, Y. Liao, J. He, D. Li, 3D printing of drug delivery systems enhanced with micro/nano-technology, *Adv. Drug Deliv. Rev.* 216 (2025) 115479.
- [69] A.R. Dixon, S.H. Jariwala, Z. Bilis, J.R. Loverde, P.F. Pasquina, L.M. Alvarez, Bridging the gap in peripheral nerve repair with 3D printed and bioprinted conduits, *Biomaterials* 186 (2018) 44–63.
- [70] T. Distler, I. Lauria, R. Detsch, C.M. Sauter, F. Bendt, J. Kapr, S. Rütten, A. R. Boccaccini, E. Fritsche, Neuronal differentiation from induced pluripotent stem cell-derived neurospheres by the application of oxidized alginate-gelatin-laminin hydrogels, *Biomedicines* 9 (3) (2021).
- [71] G.C.W. de Ruiter, M.J.A. Malesky, M.J. Yaszemski, A.J. Windebank, R.J. Spinner, Designing ideal conduits for peripheral nerve repair, *Neurosurg. Focus* 26 (2) (2009).
- [72] O. Alluin, C. Wittmann, T. Marqueste, J.-F. Chabas, S. Garcia, M.-N. Lavaut, D. Guinard, F. Feron, P. Decherchi, Functional recovery after peripheral nerve injury and implantation of a collagen guide, *Biomaterials* 30 (3) (2009) 363–373.
- [73] J. Kapr, L. Petersilie, T. Distler, I. Lauria, F. Bendt, C.M. Sauter, A.R. Boccaccini, C.R. Rose, E. Fritsche, Human induced pluripotent stem cell-derived neural progenitor cells produce distinct neural 3D *in vitro* models depending on alginate/Gellan Gum/Laminin Hydrogel blend properties, *Adv. Healthc. Mater.* 10 (16) (2021) 2100131.
- [74] K. Jansen, J.F.A. van der Werff, P.B. van Wachem, J.P.A. Nicolai, L.F.M.H. de Leij, M.J.A. van Luyn, A hyaluronan-based nerve guide: *in vitro* cytotoxicity, subcutaneous tissue reactions, and degradation in the rat, *Biomaterials* 25 (3) (2004) 483–489.
- [75] L. Vogt, F. Ruther, S. Salehi, A.R. Boccaccini, Poly(Glycerol Sebacate) in Biomedical Applications—A review of the recent literature, *Adv. Healthc. Mater.* 10 (9) (2021) 2002026.
- [76] S.J. Tan, L. Yobas, G.Y. Lee, C.N. Ong, C.T. Lim, Microdevice for the isolation and enumeration of cancer cells from blood, *Biomed. MicroDevices* 11 (4) (2009) 883–892.
- [77] F.-Y. Hsieh, H.-H. Lin, S.-h. Hsu, 3D bioprinting of neural stem cell-laden thermoresponsive biodegradable polyurethane hydrogel and potential in central nervous system repair, *Biomaterials* 71 (2015) 48–57.
- [78] G. Verreck, I. Chun, Y. Li, R. Kataria, Q. Zhang, J. Rosenblatt, A. Decorte, K. Heymans, J. Adriaensen, M. Bruining, M. Van Remoortere, H. Borghys, T. Meert, J. Peeters, M.E. Brewster, Preparation and physicochemical characterization of biodegradable nerve guides containing the nerve growth agent sabeluzole, *Biomaterials* 26 (11) (2005) 1307–1315.
- [79] A.M. Parr, D. Joung, V. Truong, C. Neitzke, S.Z. Guo, P. Walsh, J. Monat, F. B. Meng, S.H. Parka, J.R. Dutton, M.C. McAlpine, 3D Printed stem-cell derived neural progenitors generate spinal cord scaffolds, *J. Neurotrauma* 35 (16) (2018) A55. A55.
- [80] X. Liu, M. Hao, Z. Chen, T. Zhang, J. Huang, J. Dai, Z. Zhang, 3D bioprinted neural tissue constructs for spinal cord injury repair, *Biomaterials* 272 (2021).
- [81] J. Wang, X. Kong, Q. Li, C. Li, H. Yu, G. Ning, Z. Xiang, Y. Liu, S. Feng, The spatial arrangement of cells in a 3D-printed biomimetic spinal cord promotes directional differentiation and repairs the motor function after spinal cord injury, *Biofabrication* 13 (4) (2021).
- [82] L. Zhou, A.C. Wolfes, Y. Li, D.C.W. Chan, H. Ko, F.G. Szele, H. Bayley, Lipid-bilayer-supported 3D printing of Human cerebral cortex cells reveals developmental interactions, *Adv. Mater.* 32 (31) (2020) e2002183.
- [83] D. Janzen, E. Bakirci, J. Faber, M. Andrade Mier, J. Hauptstein, A. Pal, L. Forster, J. Hazur, A.R. Boccaccini, R. Detsch, J. Teßmar, S. Budday, T. Blunk, P.D. Dalton, C. Villmann, Reinforced hyaluronic acid-based matrices promote 3D neuronal network formation, *Adv. Healthc. Mater.* 11 (21) (2022) 2201826.
- [84] I. Salmon, S. Grebenyuk, A.R.A. Fattah, G. Rustandi, T. Pilkington, C. Verfaillie, A. Ranga, Engineering neurovascular organoids with 3D printed microfluidic chips, *Lab. Chip.* 22 (8) (2022) 1615–1629.
- [85] H.W. Cai, Z. Ao, C.H. Tian, Z.H. Wu, C. Kaurich, Z. Chen, M.X. Gu, A. G. Hohmann, K. Mackie, F. Guo, Engineering human spinal microphysiological systems to model opioid-induced tolerance, *Bioact. Mater.* 22 (2023) 482–490.
- [86] P.A. Goldstein, A.M. Dolga, R. Gosens, Advanced modeling of peripheral neuro-effector communication and -plasticity, *Physiology* 35 (5) (2020) 348–357.
- [87] M. Xia, J. Ma, M. Wu, L. Guo, Y. Chen, G.L. Li, S. Sun, R. Chai, H. Li, W. Li, Generation of innervated cochlear organoid recapitulates early development of auditory unit, *Stem Cell Rep.* 18 (1) (2023) 319–336.
- [88] D. Xiao, Q. Deng, Y. Guo, X. Huang, M. Zou, J. Zhong, P. Rao, Z. Xu, Y. Liu, Y. Hu, Y. Shen, K. Jin, M. Xiang, Generation of self-organized sensory ganglion organoids and retinal ganglion cells from fibroblasts, *Sci. Adv.* 6 (22) (2020) eaaz5858.
- [89] T. Lu, M. Wang, W. Zhou, Q. Ni, Y. Yue, W. Wang, Y. Shi, Z. Liu, C. Li, B. Hong, X. Zhou, S. Zhong, K. Wang, B. Zeng, J. Zhang, W. Wang, X. Zhang, Q. Wu, X. Wang, Decoding transcriptional identity in developing human sensory neurons and organoid modeling, *Cell* 187 (26) (2024) 7374–7393, e28.
- [90] A.F. Rockel, N. Wagner, P. Spenger, S. Ergün, P. Wörsdörfer, Neuro-mesodermal assemblages recapitulate aspects of peripheral nervous system development *in vitro*, *Stem Cell Reports* 18 (5) (2023) 1155–1165.
- [91] L. George, H. Dunkel, B.J. Hunnicutt, M. Filla, C. Little, R. Lansford, F. Lefcort, *In vivo* time-lapse imaging reveals extensive neural crest and endothelial cell interactions during neural crest migration and formation of the dorsal root and sympathetic ganglia, *Dev. Biol.* 413 (1) (2016) 70–85.
- [92] W.A. Anderson, A. Bosak, H.T. Hogberg, T. Hartung, M.J. Moore, Advances in 3D neuronal microphysiological systems: towards a functional nervous system on a chip, *in vitro cellular & developmental biology, Animal* 57 (2021) 191–206.
- [93] J. Kawada, S. Kaneda, T. Kirihaara, A. Maroof, T. Levi, K. Eggan, T. Fujii, Y. Ikeuchi, Generation of a motor nerve organoid with Human stem cell-derived neurons, *Stem Cell Reports* 9 (5) (2017) 1441–1449.
- [94] X.M. Spijkers, S. Pasteuning-Vuhman, J.C. Dorleijn, P. Vulto, N.R. Wevers, R. J. Pasterkamp, A directional 3D neurite outgrowth model for studying motor axon biology and disease, *Sci. Rep.* 11 (1) (2021) 2080.
- [95] D. Lee, H.Q. Tran, A.T. Dudley, K. Yang, Z. Yan, J. Xie, Advancing nerve regeneration: peripheral nerve injury (PNI) chip empowering high-speed biomaterial and drug screening, *Chem. Eng. J.* 486 (2024) 150210.
- [96] Y. Yu, B. Jin, J. Chen, C. Lou, J. Guo, C. Yang, Y. Zhao, Nerve-on-a-chip derived biomimicking microfibers for peripheral Nerve regeneration, *Adv. Sci.* 10 (20) (2023) e2207536.
- [97] S. Hyung, S.R. Lee, J. Kim, Y. Kim, S. Kim, H.N. Kim, N.L. Jeon, A 3D disease and regeneration model of peripheral nervous system-on-a-chip, *Sci. Adv.* 7 (5) (2021) 12.
- [98] K.-A. Nave, H.B. Werner, Myelination of the nervous system: mechanisms and functions, *Annu. Rev. Cell Dev. Biol.* 30 (2014) 503–533.
- [99] J. Salzer, B. Zalc, Myelination, *Curr. Biol.* 26 (20) (2016) R971–R975.
- [100] C.A. Teixeira, C.O. Miranda, V.F. Sousa, T.E. Santos, A.R. Malheiro, M. Solomon, G.H. Maegawa, P. Brites, M.M. Sousa, Early axonal loss accompanied by impaired endocytosis, abnormal axonal transport, and decreased microtubule stability occur in the model of Krabbe's disease, *Neurobiol. Dis.* 66 (2014) 92–103.
- [101] M.M. Dimachkie, R.J. Barohn, J. Katz, Multifocal Motor neuropathy, Multifocal acquired demyelinating sensory and Motor neuropathy, and other chronic acquired demyelinating polyneuropathy variants, *Neurol. Clin.* 31 (2) (2013) 533. +.
- [102] S. Hyung, S.R. Lee, J. Kim, Y. Kim, S. Kim, H.N. Kim, N.L. Jeon, A 3D disease and regeneration model of peripheral nervous system-on-a-chip, *Sci. Adv.* 7 (5) (2021).
- [103] E. De Vitis, V. La Pesa, F. Gervaso, A. Romano, A. Quattrini, G. Gigli, L. Moroni, A. Polini, A microfabricated multi-compartment device for neuron and schwann cell differentiation, *Sci. Rep.* 11 (1) (2021) 12.
- [104] J.J. Gentry, P.A. Barker, B.D. Carter, The p75 neurotrophin receptor: multiple interactors and numerous functions, in: L. Aloe, L. Calza (Eds.), *Ngf and Related Molecules in Health and Disease*, Elsevier Science Bv, Amsterdam, 2004, pp. 25–39.
- [105] H.U. Lee, S. Nag, A. Blasiak, Y. Jin, N. Thakor, I.H. Yang, Subcellular optogenetic stimulation for activity-dependent myelination of axons in a novel microfluidic compartmentalized platform, *ACS. Chem. Neurosci.* 7 (10) (2016) 1317–1324.
- [106] A.D. Sharma, L. McCoy, E. Jacobs, H. Willey, J.Q. Behn, H. Nguyen, B. Bolon, J. L. Curley, M.J. Moore, Engineering a 3D functional human peripheral nerve *in vitro* using the Nerve-on-a-Chip platform, *Sci. Rep.* 9 (2019) 12.
- [107] H.S. Park, S. Liu, J. McDonald, N. Thakor, I.H. Yang, Neuromuscular junction in a microfluidic device, in: 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE, 2013, pp. 2833–2835.
- [108] S.G.M. Uzel, R.J. Platt, V. Subramanian, T.M. Pearl, C.J. Rowlands, V. Chan, L. A. Boyer, P.T.C. So, R.D. Kamm, Microfluidic device for the formation of optically excitable, three-dimensional, compartmentalized motor units, *Sci. Adv.* 2 (8) (2016) 13.
- [109] O.F. Vila, S.G.M. Uzel, S.P. Ma, D. Williams, J. Pak, R.D. Kamm, G. Vunjak-Novakovic, Quantification of human neuromuscular function through optogenetics, *Theranostics* 9 (5) (2019) 1232–1246.
- [110] K.S. Dittlau, E.N. Krasnow, L. Fumagalli, T. Vandormer, P. Baatsen, A. Kerstens, G. Giacomazzi, B. Pavie, E. Rossaert, J. Beckers, M. Sampaoli, P. Van Damme, L. Van den Bosch, Human motor units in microfluidic devices are impaired by *FUS* mutations and improved by HDAC6 inhibition, *Stem Cell Rep.* 16 (9) (2021) 2213–2227.
- [111] J. Sala-Jarque, F. Mesquida-Veny, M. Badiola-Mateos, J. Samitier, A. Hervera, J. A. del Río, Neuromuscular activity induces paracrine signaling and triggers axonal regrowth after injury in microfluidic lab-on-chip devices, *Cells* 9 (2) (2020) 21.
- [112] D. Zhu, A. Nilghaz, Z. Tong, D.P. Poole, K. O'Sullivan, W.L. Imlach, R. V. Haberberger, N.A. Veldhuis, D. Matusica, N.H. Voelcker, Pain-on-a-chip: a microfluidic device for neuron differentiation and functional discrimination in animal models of chronic Pain, *Biosens. Bioelectr.* (2025) 117401.
- [113] L.K. Vora, A.D. Gholap, K. Jetha, R.R.S. Thakur, H.K. Solanki, V.P. Chavda, Artificial intelligence in pharmaceutical technology and drug delivery design, *Pharmaceutics* 15 (7) (2023).

- [114] J. Chen, D. Huang, L. Wang, J. Hou, H. Zhang, Y. Li, S. Zhong, Y. Wang, Y. Wu, W. Huang, 3D bioprinted multiscale composite scaffolds based on gelatin methacryloyl (GelMA)/chitosan microspheres as a modular bioink for enhancing 3D neurite outgrowth and elongation, *J. Colloid. Interface Sci.* 574 (2020) 162–173.
- [115] A. Malheiro, F. Morgan, M. Baker, L. Moroni, P. Wieringa, A three-dimensional biomimetic peripheral nerve model for drug testing and disease modelling, *Biomaterials* 257 (2020) 120230.
- [116] A. Malheiro, A. Harichandan, J. Bernardi, A. Seijas-Gamardo, G.F. Konings, P.G. A. Volders, A. Romano, C. Mota, P. Wieringa, L. Moroni, 3D culture platform of human iPSCs-derived nociceptors for peripheral nerve modeling and tissue innervation, *Biofabrication*. 14 (1) (2022) 21.
- [117] E.G.Z. Centeno, H. Cimarosti, A. Bithell, 2D versus 3D human induced pluripotent stem cell-derived cultures for neurodegenerative disease modelling, *Mol. Neurodegener.* 13 (2018) 15.
- [118] A. Ruiz, P. Joshi, R. Mastrangelo, M. Francolini, C. Verderio, M. Matteoli, Testing a toxicity on primary CNS cultures using drug-screening microfluidic chips, *Lab. Chip.* 14 (15) (2014) 2860–2866.
- [119] M. Jang, N. Choi, H.N. Kim, Hyperglycemic neurovasculature-on-A-chip to study the effect of SIRT1-targeted therapy for the type 3 diabetes "Alzheimer's Disease", *Adv. Sci.* 9 (34) (2022) 13.
- [120] C. Zivko, R. Sagar, A. Xydia, A. Lopez-Montes, J. Mintzer, P.B. Rosenberg, D. M. Shade, A.P. Porsteinsson, C.G. Lyketsov, V. Mahairaki, iPSC-derived hindbrain organoids to evaluate escitalopram oxalate treatment responses targeting neuropsychiatric symptoms in Alzheimer's disease, *Mol. Psychiatry* 29 (11) (2024) 3644–3652.
- [121] S. Bolognin, M. Fossépré, X.B. Qing, J. Jarazo, J. Scancar, E.L. Moreno, S. L. Nickels, K. Wasner, N. Ouzren, J. Walter, A. Grünwald, E. Glaab, L. Salamanca, R.M.T. Fleming, P.M.A. Antony, J.C. Schwamborn, 3D Cultures of Parkinson's Disease-specific dopaminergic neurons for high content phenotyping and drug testing, *Adv. Sci.* 6 (1) (2019) 14.
- [122] J.J. Metzger, C. Pereda, A. Adhikari, T. Harembaki, S. Galgoczi, E.D. Siggia, A. H. Brivanlou, F. Etoc, Deep-learning analysis of micropattern-based organoids enables high-throughput drug screening of Huntington's disease models, *Cell Rep. Methods* 2 (9) (2022) 14.
- [123] T. Osaki, S.G.M. Uzel, R.D. Kamm, Microphysiological 3D model of amyotrophic lateral sclerosis (ALS) from human iPS-derived muscle cells and optogenetic motor neurons, *Sci. Adv.* 4 (10) (2018) 15.
- [124] W.W. Xue, B. Li, H.H. Liu, Y.J. Xiao, B. Li, L. Ren, H.J. Li, Z.C. Shao, Generation of dorsoventral human spinal cord organoids via functionalizing composite scaffold for drug testing, *iScience* 26 (1) (2023) 21.
- [125] B. Deleglise, B. Lassus, V. Soubeyre, A. Alleaume-Butaux, J.J. Hjorth, M. Vignes, B. Schneider, B. Brugg, J.L. Viowy, J.M. Peyrin, Synapto-protective drugs evaluation in reconstructed neuronal network, *PLoS. One* 8 (8) (2013) 9.
- [126] Z. Wang, M.D. Wood, S.E. Mackinnon, S. Sakiyama-Elbert, A microfluidic platform to study the effects of GDNF on neuronal axon entrapment, *Tissue Eng. Part A* 22 (2016) S152–S153.
- [127] S.N. Wang, Z. Wang, X.Y. Wang, X.P. Zhang, T.Y. Xu, C.Y. Miao, Humanized cerebral organoids-based ischemic stroke model for discovering of potential anti-stroke agents, *Acta Pharmacol. Sin.* 44 (3) (2023) 513–523.
- [128] W. Zhu, Y. Wu, X. Li, H. Yang, F. He, J. Ma, J. Wei, L. Leng, A stroke organoids-multomics platform to study injury mechanism and drug response, *Bioact. Mater.* 44 (2025) 68–81.
- [129] R. Cecchelli, S. Aday, E. Sevin, C. Almeida, M. Culot, L. Dehouck, C. Coisne, B. Engelhardt, M.P. Dehouck, L. Ferreira, A stable and reproducible Human blood-brain barrier model derived from hematopoietic stem cells, *PLoS. One* 9 (6) (2014) 11.
- [130] Y.T. Fan, D.T. Nguyen, Y. Akay, F. Xu, M. Akay, Engineering a brain cancer chip for high-throughput drug screening, *Sci. Rep.* 6 (2016) 12.
- [131] S. Plummer, S. Wallace, G. Ball, R. Lloyd, P. Schiapparelli, A. Quiñones-Hinojosa, T. Hartung, D. Pamies, A Human iPSC-derived 3D platform using primary brain cancer cells to study drug development and personalized medicine, *Sci. Rep.* 9 (2019) 11.
- [132] A. Ramani, G. Pasquini, N.J. Gerkau, V. Jadhav, O.S. Vinchure, N. Altinisk, H. Windoffer, S. Muller, I. Rothenaigner, S. Lin, A. Mariappan, D. Rathinam, A. Mirsaidi, O. Goureau, L. Ricci-Vitiani, Q.G. D'Alessandris, B. Wollnik, A. Muotri, L. Freifeld, N. Jurisch-Yaksi, R. Pallini, C.R. Rose, V. Busskamp, E. Gabriel, K. Hadian, J. Gopalakrishnan, Reliability of high-quantity human brain organoids for modeling microcephaly, glioma invasion and drug screening, *Nat. Commun.* 15 (1) (2024).
- [133] N.J. Abbott, in: Physiology of the blood-brain barrier and its consequences for drug transport to the brain, 11th Drug Transporters and Diseased Brain Conference, Elsevier Science Bv, Girona, Spain, 2004, pp. 3–18.
- [134] E. Jagtiani, M. Yeolekar, S. Naik, V. Patravale, *In vitro* blood brain barrier models: an overview, *J. Controlled Release* 343 (2022) 13–30.
- [135] I. Nagy, B. Tóth, Z. Gáborik, F. Erdo, P. Krajcsi, Membrane transporters in physiological barriers of pharmacological importance, *Curr. Pharm. Des.* 22 (35) (2016) 5347–5372.
- [136] C. Ma, Y.S. Peng, H.T. Li, W.Q. Chen, Organ-on-a-chip: a new paradigm for drug development, *Trends. Pharmacol. Sci.* 42 (2) (2021) 119–133.
- [137] Y. Koo, B.T. Hawkins, Y. Yun, Three-dimensional (3D) tetra-culture brain on chip platform for organophosphate toxicity screening, *Sci. Rep.* 8 (2018) 7.
- [138] L. Verneti, A. Gough, N. Baetz, S. Blutt, J.R. Broughman, J.A. Brown, J. Foulke-Abel, N. Hasan, J. In, E. Kelly, O. Kovbasnjuk, J. Repper, N. Senutovitch, J. Stabb, C. Yeung, N.C. Zachos, M. Donowitz, M. Estes, J. Himmelfarb, G. Truskey, J. P. Wikswo, D.L. Taylor, Functional coupling of Human microphysiology systems: intestine, liver, kidney proximal tubule, blood-brain barrier and skeletal muscle, *Sci. Rep.* 7 (1) (2017) 42296.
- [139] B.M. Maoz, A. Herland, E.A. FitzGerald, T. Grevesse, C. Vidoudez, A.R. Pacheco, S.P. Sheehy, T.-E. Park, S. Dauth, R. Mannix, N. Budnik, K. Shores, A. Cho, J. C. Nawroth, D. Segrè, B. Budnik, D.E. Ingber, K.K. Parker, A linked organ-on-chip model of the human neurovascular unit reveals the metabolic coupling of endothelial and neuronal cells, *Nat. Biotechnol.* 36 (9) (2018) 865–874.
- [140] H. Kimura, M. Nishikawa, N. Kutsuzawa, F. Tokito, T. Kobayashi, D. A. Kurniawan, H. Shioda, W. Cao, K. Shinha, H. Nakamura, K. Doi, Y. Sakai, Advancements in microphysiological systems: exploring organoids and organ-on-a-chip technologies in drug development - focus on pharmacokinetics related organs, *Drug Metab. Pharmacokinet.* (2024) 101046.
- [141] L. Verneti, A. Gough, N. Baetz, S. Blutt, J.R. Broughman, J.A. Brown, J. Foulke-Abel, N. Hasan, J. In, E. Kelly, O. Kovbasnjuk, J. Repper, N. Senutovitch, J. Stabb, C. Yeung, N.C. Zachos, M. Donowitz, M. Estes, J. Himmelfarb, G. Truskey, J. P. Wikswo, D.L. Taylor, Functional coupling of Human microphysiology systems: intestine, liver, kidney proximal tubule, blood-brain barrier and skeletal muscle, *Sci. Rep.* 7 (1) (2017) 42296.
- [142] Z. Li, D. Li, Y. Guo, Y. Wang, W. Su, Evaluation of hepatic drug-metabolism for glioblastoma using liver-brain chip, *Biotechnol. Lett.* 43 (2) (2021) 383–392.
- [143] A. Herland, B. Maoz, D. Das, M.R. Somayaji, R. Prantil-Baun, R. Novak, M. Cronce, T. Huffstater, S.S.F. Jeanty, M. Ingram, A. Chalkiadaki, D.B. Chou, S. Marquez, A. Delahanty, S. Jallili-Firoozinezhad, Y. Milton, A. Sontheimer-Phelps, B. Swenor, O. Levy, K.K. Parker, A. Przekwas, D.E. Ingber, Quantitative prediction of human pharmacokinetic responses to drugs via fluidically coupled vascularized organ chips, *Nat. Biomed. Eng.* (2020) 19.
- [144] C.E. Staicu, F. Jipa, E. Axente, M. Radu, B.M. Radu, F. Sima, Lab-on-a-chip platforms as tools for drug screening in neuropathologies associated with blood-brain barrier alterations, *Biomolecules* 11 (6) (2021) 23.
- [145] A. Gramowski, J. Flossdorf, K. Bhattacharya, L. Jonas, M. Lantow, Q. Rahman, D. Schifmann, D.G. Weiss, E. Dopp, Nanoparticles induce changes of the electrical activity of neuronal networks on microelectrode array neurochips, *Environ. Health Perspect.* 118 (10) (2010) 1363–1369.
- [146] G.J. Nierode, B.C. Perea, S.K. McFarland, J.F. Pascoal, D.S. Clark, D.V. Schaffer, J. S. Dordick, High-throughput toxicity and phenotypic screening of 3D Human neural progenitor cell cultures on a microarray chip platform, *Stem Cell Rep.* 7 (5) (2016) 970–982.
- [147] T.H. Kim, W.A. El-Said, J.W. Choi, Highly sensitive electrochemical detection of potential cytotoxicity of CdSe/ZnS quantum dots using neural cell chip, *Biosens. Bioelectron.* 32 (1) (2012) 266–272.
- [148] Y.Q. Wang, L. Wang, Y.J. Zhu, J.H. Qin, Human brain organoid-on-a-chip to model prenatal nicotine exposure, *Lab. Chip.* 18 (6) (2018) 851–860.
- [149] S. Hong, J. Lee, Y. Kim, E. Kim, K. Shin, AAVS1-targeted, stable expression of ChR2 in human brain organoids for consistent optogenetic control, *Bioeng. Transl. Med.* 9 (6) (2024) e10690.
- [150] M. Pérez-Aliacar, M.H. Doweidar, M. Doblaré, J. Ayensa-Jiménez, Predicting cell behaviour parameters from glioblastoma on a chip images. A deep learning approach, *Comput. Biol. Med.* 135 (2021) 104547.
- [151] A.-N. Cho, Y. Jin, Y. An, J. Kim, Y.S. Choi, J.S. Lee, J. Kim, W.-Y. Choi, D.-J. Koo, W. Yu, G.-E. Chang, D.-Y. Kim, S.-H. Jo, J. Kim, S.-Y. Kim, Y.-G. Kim, J.Y. Kim, N. Choi, E. Cheong, Y.-J. Kim, H.S. Je, H.-C. Kang, S.-W. Cho, Microfluidic device with brain extracellular matrix promotes structural and functional maturation of human brain organoids, *Nat. Commun.* 12 (1) (2021) 4730.
- [152] Z. Wang, Y. Zhang, Z. Li, H. Wang, N. Li, Y. Deng, Microfluidic brain-on-a-chip: from key technology to system integration and application, *Small* 19 (52) (2023) 2304427.
- [153] N. Tabatabaei Rezaei, H. Kumar, H. Liu, S.S. Lee, S.S. Park, K. Kim, Recent advances in organ-on-chips integrated with bioprinting technologies for drug screening, *Adv. Healthc. Mater.* 12 (20) (2023) 2203172.
- [154] H. Li, H. Zhou, C. Xu, Y. Wei, X. Tang, 3D bioprinting and scaffold-free strategies for fabrication of multi-cellular tissues or organoids, *IJB* 9 (6) (2023).
- [155] B.Q. Lai, B. Feng, M.T. Che, L.J. Wang, S. Cai, M.Y. Huang, H.Y. Gu, B. Jiang, E. A. Ling, M. Li, X. Zeng, Y.S. Zeng, A modular assembly of spinal cord-like tissue allows targeted tissue repair in the transected spinal cord, *Adv. Sci.* 5 (9) (2018) 1800261.
- [156] J.-H. Lee, H. Shin, M.R. Shaker, H.J. Kim, S.-H. Park, J.H. Kim, N. Lee, M. Kang, S. Cho, T.H. Kwak, J.W. Kim, M.-R. Song, S.-H. Kwon, D.W. Han, S. Lee, S.-Y. Choi, I.J. Rhyu, H. Kim, D. Geum, I.-J. Cho, W. Sun, Production of human spinal-cord organoids recapitulating neural-tube morphogenesis, *Nat. Biomed. Eng.* 6 (4) (2022) 435–445.
- [157] Y. Morofuji, S. Nakagawa, Drug development for Central nervous system diseases using *In vitro* blood-brain barrier models and Drug repositioning, *Curr. Pharm. Des.* 26 (13) (2020) 1466–1485.
- [158] C. Probst, S. Schneider, P. Loskill, High-throughput organ-on-a-chip systems: current status and remaining challenges, *Curr. Opin. Biomed. Eng.* 6 (2018) 33–41.
- [159] O. Frey, P.M. Misun, D.A. Fluri, J.G. Hengstler, A. Hierlemann, Reconfigurable microfluidic hanging drop network for multi-tissue interaction and analysis, *Nat. Commun.* 5 (1) (2014) 4250.
- [160] D.T.T. Phan, X. Wang, B.M. Craver, A. Sobrino, D. Zhao, J.C. Chen, L.Y.N. Lee, S. C. George, A.P. Lee, C.C.W. Hughes, A vascularized and perfused organ-on-a-chip platform for large-scale drug screening applications, *Lab. Chip.* 17 (3) (2017) 511–520.
- [161] S.J. Trietsch, E. Naumovska, D. Kurek, M.C. Setyawati, M.K. Vormann, K. J. Wilschut, H.L. Lanz, A. Nicolas, C.P. Ng, J. Joore, S. Kustermann, A. Roth, T. Hankemeier, A. Moisan, P. Vulto, Membrane-free culture and real-time barrier

- integrity assessment of perfused intestinal epithelium tubes, *Nat. Commun.* 8 (1) (2017) 262.
- [162] S.W. Deng, C.F. Li, J.X. Cao, Z. Cui, J. Du, Z. Fu, H.J. Yang, P. Chen, Organ-on-a-chip meets artificial intelligence in drug evaluation, *Theranostics* 13 (13) (2023) 4526–4558.
- [163] K. Paek, S. Kim, S. Tak, M.K. Kim, J. Park, S. Chung, T.H. Park, J.A. Kim, A high-throughput biomimetic bone-on-a-chip platform with artificial intelligence-assisted image analysis for osteoporosis drug testing, *Bioeng. Transl. Med.* 8 (1) (2023) 14.
- [164] E.W. Esch, A. Bahinski, D. Huh, Organs-on-chips at the frontiers of drug discovery, *Nat. Rev. Drug Discovery* 14 (4) (2015) 248–260.